# Evaluation orders on bracketings in the hyperoperation hierarchy

Toshihisa Urahigashi
Independent researcher, Nara, Japan
123026.urahigashi.toshihisa@gmail.com
ORCID 0009-0004-0460-6242

**Abstract**

Let $\mathrm{H}_r$ be the Goodstein hyperoperation hierarchy, so that $\mathrm{H}_3$ is exponentiation and $\mathrm{H}_4$ tetration. For $r \geq 3$ the operation is not associative, so the value of $\mathrm{H}_r(a_1, \ldots, a_n)$ depends on the bracketing, that is on a binary tree with $n$ leaves. Write $T \leq_{\mathrm{ev}} U$ when $\mathrm{ev}_r(T) \leq \mathrm{ev}_r(U)$ for *every* assignment of leaf values $\geq 2$, and let $\leq_{\mathrm{St}}$ be the Stanley order on those trees, which extends the Tamari order.

At rank 3 we determine $\leq_{\mathrm{ev}}$ completely: it is the Stanley order. The inclusion $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ is due to Chow; we prove its converse, which he left open, constructively and with a bound. If $T \not\leq_{\mathrm{St}} U$ then a *single* leaf separates them, and the value to put there is written down in advance as $M = 2K^2$, where $K$ is a product of subtree values read off from $U$. One family of $n-2$ assignments, given in closed form, therefore decides the entire order.

For every $r \geq 4$ we prove that $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$, the value increasing strictly along every Stanley cover; the semi-associative law that underlies this is not new, and we say where it is already stated. Equality in that law is settled by one statement covering every rank and every boundary case. The device throughout is the vector of right depths of the leaves. Much of the paper is machine-checked in Lean 4, and it is explicit about what is not.

**Keywords** Stanley order · Tamari lattice · Dyck paths · Hyperoperations · Iterated exponentiation · Formal verification

## Acknowledgements

The rank-three question is Spivak's. The inclusion $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ is Chow's, and Rubey independently reduced it to a single inequality; we give a second proof of it in Section 7 but the result is Chow's. The suggestion that a single enlarged variable should do the work is Chow's as well, and Spivak's question already contains the smallest instance of it; Rubey sketched how the converse might be reduced to a simpler comparison. We thank the MathOverflow discussion [19] for all of this.

## 1 Introduction

### 1.1 The question

Exponentiation is not associative, so the $C_{n-1}$ bracketings of $a_1 \star \cdots \star a_n$ can take different values, and one may ask how those values are *arranged*. Two orders present themselves. The bracketings are the binary trees with $n$ leaves; they carry the Tamari lattice, generated by

right rotations, and, under the standard bijection with Dyck paths, the Stanley order, in which one path lies weakly below another. Against these stands the arithmetic order: write $T \leq_{\mathrm{ev}} U$ when $\mathrm{ev}_r(T) \leq \mathrm{ev}_r(U)$ holds *for every* assignment of values $\geq 2$ to the leaves.

Spivak [19] asked, for $r = 3$, whether $\leq_{\mathrm{ev}}$ and the Stanley order agree. Chow, following a comment of Rubey, proved the inclusion $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$, writing that he could "complete half of the proof"; Rubey, independently, reduced that inclusion to a single inequality and stopped there, writing "it remains to check that…". Chow decomposed the whole problem into that inclusion together with its converse, and left the converse — that bracketings which are *not* Stanley-related can be made to compare either way by a suitable choice of leaf values — open. Spivak's question already contains the smallest instance of that converse: "To see that $C$ and $D$ are incomparable: use $a = b = c = 2$ and $d =$ large to obtain $C > D$; and use $a = b = d = 2$ and $c =$ large to obtain $D > C$." What is missing there, and what Theorem 1.2 supplies, is how large "large" has to be, and that one leaf suffices at every $n$. The expected answer was not obvious: Stanley, commenting on the question, pointed to a paper of Stembridge on a different order attached to iterated exponentials and remarked that it "lends credence to the possibility that the iterated exponentiation order is not the Dyck order" [19].

Above rank 3 the literature compares the bracketings of three arguments (Section 17), which is enough to relate evaluation to the Tamari order; the Stanley order, as far as we know, had not been related to evaluation there at all. Theorem 14.3 below supplies one of the two inclusions, and the converse remains open. The behaviour of the relation with the Tamari order changes with the rank in a way that can be pinned down exactly.

### 1.2 Plan

The paper has two parts. Part I (Sections 3–9) settles rank 3: the evaluation order *is* the Stanley order, with an explicit separating assignment and a finite test family. Part II (Sections 10–14) goes up the hierarchy: the semi-associative law at every rank, the complete classification of its equality cases, the resulting dichotomy between rank 3 and rank $\geq 4$, and strict Stanley monotonicity at every higher rank. Sections 15 and 16 describe the computations and the Lean development, Section 17 the attribution, and Section 18 what is left.

### 1.3 Results

Throughout, $\mathrm{H}_r$ is the hyperoperation hierarchy fixed in Section 2, with $\mathrm{H}_1(a,b) = a + b$, $\mathrm{H}_2(a,b) = ab$, $\mathrm{H}_3(a,b) = a^b$ and $\mathrm{H}_4(a,b) = a \uparrow\uparrow b$; $\mathcal{T}_n$ is the set of binary trees with $n$ leaves, and $\mathrm{ev}_r$ their evaluation at leaf values $\geq 2$.

**Part I: rank three.** Here the answer is complete.

**Theorem 1.1** (Rank three: the evaluation order)**.** *For every $n$ and all $T, U \in \mathcal{T}_n$,*

$$T \leq_{\mathrm{ev}} U \quad \Longleftrightarrow \quad T \leq_{\mathrm{St}} U.$$

The proof is constructive, and the witness it produces is unexpectedly simple: one leaf does all the work, and the value put there is written down in advance.

**Theorem 1.2** (Separation by a single leaf)**.** *Let $T, U \in \mathcal{T}_n$ with $T \not\leq_{\mathrm{St}} U$, and let $i$ be an index with $r_i(T) > r_i(U)$ (Section 4 shows there is one). Put $K = K_i(U)$ as in Definition 6.1 and $M = 2K^2$. Then, for the assignment that gives leaf $i$ the value $M$ and every other leaf the value 2,*

$$\mathrm{ev}_3(T) > \mathrm{ev}_3(U).$$

Because the value $M$ can be bounded uniformly over all trees with $n$ leaves, a single finite family of assignments decides the order.

**Corollary 1.3** (A finite test family)**.** *Let $n \geq 2$, let $F_0(t) = t$ and $F_{h+1}(t) = 2^{F_h(t)}$, and put $M_n = 2F_{n-2}(2)^2$. For $3 \leq i \leq n$ let $\vec{a}^{(i)}$ be the assignment with leaf $i$ equal to $M_n$ and every other leaf equal to 2. Then for all $T, U \in \mathcal{T}_n$,*

$$T \leq_{\mathrm{St}} U \quad \Longleftrightarrow \quad \mathrm{ev}_3(T)(\vec{a}^{(i)}) \leq \mathrm{ev}_3(U)(\vec{a}^{(i)}) \ \text{ for } i = 3, \ldots, n.$$

So an order defined by a universal quantifier over infinitely many assignments is decided by $n-2$ of them, given in closed form. The first two leaves may be omitted because $r_1(T) = 0$ and $r_2(T) = 1$ for *every* $T \in \mathcal{T}_n$ (Lemma 2.7), so those two coordinates never distinguish two trees.

**Part II: up the hierarchy.** Above rank 3 we do not know the analogue of Theorem 1.1; what survives, and can be made exact, are the relations with the Tamari and Stanley orders.

**Theorem 1.4** (Rank three: the local law)**.** *Let $x \geq 1$ and $y \geq 2$. Then $(x^y)^z \leq x^{(y^z)}$ for every $z \geq 0$. If moreover $x \geq 2$, equality holds if and only if $z = 1$ or $(y, z) = (2, 2)$. The hypothesis $y \geq 2$ cannot be dropped: for $y = 1$ and all $x, z \geq 2$ the inequality fails.*

**Theorem 1.5** (Rank 4)**.** *Let $x \geq 2$ and $y \geq 2$. Then $(x \uparrow\uparrow y) \uparrow\uparrow z \leq x \uparrow\uparrow (y \uparrow\uparrow z)$ for every $z \geq 0$. If moreover $z \geq 2$, the inequality is strict.*

**Proposition 1.6** (Tamari monotonicity)**.** *Let $a \geq 2$ and $r \geq 3$. If $U$ arises from $T$ by a right rotation at some node (these rotations are exactly the covers $T \lessdot U$ of the Tamari lattice), then $\mathrm{ev}_{r,a}(T) \leq \mathrm{ev}_{r,a}(U)$. Consequently $\mathrm{ev}_{r,a}$ is order-preserving for the Tamari order. Moreover, for the right comb $R_n$ with $n$ leaves, $\mathrm{ev}_{r,a}(R_n) = \mathrm{H}_{r+1}(a, n)$.*

**Theorem 1.7** (Strict Tamari monotonicity from rank four)**.** *Let $a \geq 2$ and $r \geq 4$. If $T \lessdot U$ is a Tamari cover, then*

$$\mathrm{ev}_{r,a}(T) < \mathrm{ev}_{r,a}(U).$$

*Thus for $r \geq 4$ the evaluation is strictly increasing along* every *cover of the Tamari lattice, not merely along root rotations.*

**Corollary 1.8** (Rank dichotomy)**.** *For $a \geq 2$ the evaluation $\mathrm{ev}_{r,a}$ is order-preserving for the Tamari order at every rank $r \geq 3$ (Proposition 1.6), and*

- *at rank 3 strictness can fail, but only for $a = 2$: the cover $((aa)a)a \lessdot (aa)(aa)$ carries equal values, both 256, when $a = 2$;*
- *at every rank $r \geq 4$ it is strict on covers (Theorem 1.7).*

*In fact $(r, a) = (3, 2)$ is the only exception: for every other pair with $r \geq 3$ and $a \geq 2$ every cover is strict (Corollary 13.2).*

**Theorem 1.9** (All ranks)**.** *For every $r \geq 3$ and all $x \geq 2$, $y \geq 2$ and $z \geq 0$,*

$$\mathrm{H}_r\big(\mathrm{H}_r(x, y), z\big) \ \leq \ \mathrm{H}_r\big(x, \mathrm{H}_r(y, z)\big).$$

**Theorem 1.10** (Complete equality classification)**.** *Let $x \geq 2$ and $y \geq 2$.*

1. *For every $r \geq 2$ the two sides are equal when $z = 1$; in particular equality holds in Theorem 1.9 for every $r \geq 3$.*
2. *For $r = 3$ the only further equality is $(y, z) = (2, 2)$.*

3. *For $r \geq 4$ and $z \geq 2$ the inequality is* strict.

4. *For every $r \geq 3$ the inequality is strict when $z = 0$, since* $\mathrm{H}_r(\mathrm{H}_r(x, y), 0) = 1 < x = \mathrm{H}_r(x, \mathrm{H}_r(y, 0))$.

*Items (1), (3) and (4) together give: for $r \geq 4$ equality holds if and only if $z = 1$.*

Theorems 1.9 and 1.10 are proved in Section 12 by inductions on the rank whose engine is the identity-like inequality ($\mathrm{SUM}_r$) below; the case $r = 3$ is Theorem 1.4 and the case $r = 4$ can also be obtained directly, as in Section 10.

**Remark 1.11** (The statement is not an asymptotic triviality)**.** Two features are worth isolating. First, ($\mathrm{HT}_r$) does not follow from monotonicity alone: on the left the *first* argument grows, $x \mapsto \mathrm{H}_r(x, y) > x$, while on the right the *second* argument grows, $z \mapsto \mathrm{H}_r(y, z)$, so the two coordinates compete. Second, high rank does not make everything large: since $\mathrm{H}_r(a, 1) = a$ for $r \geq 2$, one gets

$$\mathrm{H}_r(2, 2) = \mathrm{H}_{r-1}(2, \mathrm{H}_r(2, 1)) = \mathrm{H}_{r-1}(2, 2) = \cdots = \mathrm{H}_2(2, 2) = 4$$

for *every* $r \geq 2$. A degenerate corner therefore survives at every rank, and it is exactly this corner that carries the rank-3 equality of Theorem 1.10(2) and that has to be treated separately in the proof of Theorem 1.10(3).

**Remark 1.12** (How the two parts fit together)**.** The Tamari order is contained in the Stanley order. At rank 3, therefore, Theorem 1.1 contains Theorem 13.1 for $r = 3$: Tamari comparability implies Stanley comparability, hence comparability in $\leq_{\mathrm{ev}}$. What Theorem 13.1 adds is the ranks $r \geq 4$, where Theorem 1.1 has no known analogue; and what Theorem 1.10 adds at rank 3 is the classification of *equality*, which an order statement does not see.

### 1.4 The mechanism, and what in it is not new

Everything turns on one vector. Let $r_i(T)$ be the number of *right* branches on the path from the root to the $i$th leaf. The vector $r(T) = (r_1(T), \ldots, r_n(T))$ does two things at once:

- it presents the Stanley order coordinatewise: $T \leq_{\mathrm{St}} U$ if and only if $r_i(T) \leq r_i(U)$ for all $i$ (Proposition 4.1);
- it counts the exponentiations through which an enlarged leaf value passes, and therefore the height of the tower that value generates (Proposition 6.3).

**Both of these are already in the literature.** Knuth defines the Stanley lattice (introduced by Stanley [21, pp. 222–223]) on $n$-node forests by declaring $F \subseteq F'$ when the depth coordinates satisfy $c_j \leq c'_j$ for every $j$ [14, §7.2.1.6, exercise 28 and Fig. 41]; the passage to binary trees is the identification of the depth of a vertex in a DFS tree with its right depth, stated explicitly by Lehtonen and Waldhauser [17, §2.5], who in turn attribute the underlying bijection to Csákány and Waldhauser [4, 2.8] ("Bracketings are uniquely determined by their right (or left) depth sequences"); the same material is restated in the Appendix of Waldhauser's dissertation [24], whose Proposition A.7 repeats the argument of [4, 3.4] for exponentiation. And the second item is the key observation of [4, 3.4]: for exponentiation, "the height of the level of $p_i$ in $b$ always equals the right depth of $x_i$ in $B$". From this Csákány and Waldhauser deduce, by substituting distinct primes and invoking unique factorisation, that distinct bracketings induce distinct operations. We give a short proof of Proposition 4.1 in Section 4 only to fix conventions and to connect the coordinates to the Dyck-path convention of [19].

The idea of enlarging a single variable is not new either: it is exactly what Chow proposes in the last paragraph of his answer as the way to attack the missing half, and Rubey, in comments under his own answer, expected it to work and sketched how the missing half might be reduced to a simpler comparison, without completing the sketch.

What we supply is the step from *difference* to *direction*. Distinct primes show that two values are unequal but say nothing about which is larger; to get the order one needs a two-sided estimate for the tower that an enlarged leaf generates, and that is Proposition 6.3. With it in hand the remaining questions — which leaf to enlarge, how far, and why every other leaf may be left at 2 — all have explicit answers.

# 2 Preliminaries

## 2.1 The hierarchy

**Definition 2.1** (Hyperoperation [7])**.** Define $\mathrm{H} : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$\mathrm{H}_0(a,b) = b+1, \quad \mathrm{H}_1(a,0) = a, \quad \mathrm{H}_2(a,0) = 0, \quad \mathrm{H}_{r+3}(a,0) = 1,$$

$$\mathrm{H}_{r+1}(a,b+1) = \mathrm{H}_r\big(a, \mathrm{H}_{r+1}(a,b)\big).$$

**Lemma 2.2.** *For all $a, b$ we have $\mathrm{H}_1(a,b) = a+b$, $\mathrm{H}_2(a,b) = ab$ and $\mathrm{H}_3(a,b) = a^b$. Moreover $\mathrm{H}_r(a,1) = a$ for $r \geq 2$.*

*Proof.* For ranks 1, 2 and 3, induction on the second argument gives respectively addition, repeated addition and repeated multiplication. The identity at second argument 1 is immediate at rank 2. For $r \geq 3$ the recursion gives $\mathrm{H}_r(a,1) = \mathrm{H}_{r-1}(a, \mathrm{H}_r(a,0)) = \mathrm{H}_{r-1}(a,1) = a$ by induction on $r$. □

The elementary growth and monotonicity consequences of this recursion are proved in Appendix B; they are used in both parts.

We write $a \uparrow\uparrow b := \mathrm{H}_4(a,b)$, so $a \uparrow\uparrow 0 = 1$ and $a \uparrow\uparrow (b+1) = a^{(a \uparrow\uparrow b)}$.

## 2.2 Bracketings and evaluation

**Definition 2.3** (Trees and evaluation)**.** $\mathcal{T}_n$ is the set of binary trees with $n$ leaves; a tree is either a leaf or an ordered pair $(L,R)$ of trees. Leaves are numbered $1, \dots, n$ from left to right. Given $\vec{a} \in \mathbb{N}_{\geq 2}^n$ we set $\mathrm{ev}_r(\text{leaf } i) = a_i$ and $\mathrm{ev}_r\big((L,R)\big) = \mathrm{H}_r\big(\mathrm{ev}_r(L), \mathrm{ev}_r(R)\big)$, and we write $\mathrm{ev}_r(T)(\vec{a})$ when the assignment is to be displayed. A tree together with an assignment is a *labelled tree*; rotation permutes no labels. When every $a_i$ equals a single value $a$ we write $\mathrm{ev}_{r,a}(T)$.

At rank 3 we abbreviate $\mathrm{ev}_3\big((L,R)\big) = \mathrm{ev}_3(L)^{\mathrm{ev}_3(R)}$, and we write $T \leq_{\mathrm{ev}} U$ when $\mathrm{ev}_3(T)(\vec{a}) \leq \mathrm{ev}_3(U)(\vec{a})$ for every $\vec{a} \in \mathbb{N}_{\geq 2}^n$; this is the *evaluation order* of Part I. In Part II the same symbol denotes the relation defined in the same way from $\mathrm{ev}_r$, at the rank $r$ under consideration, which is always stated; above rank 3 it is a priori only a preorder, and nothing below uses its antisymmetry. Every value is at least 2, since the leaves are and $\mathrm{H}_r(x,y) \geq x \geq 2$ for $x, y \geq 2$ and $r \geq 2$; this is used without further comment.

## 2.3 The two orders

**Definition 2.4** (Right rotation and the Tamari order)**.** The *right rotation at the root* sends $((A \cdot B) \cdot C)$ to $(A \cdot (B \cdot C))$. The Tamari order on $\mathcal{T}_n$ is generated by right rotations performed at arbitrary nodes; that this order is a lattice, with the left comb as least and the right comb

as greatest element, is the theorem of Friedman and Tamari [5], whose title already names the relation a *loi demi-associative*; for its proof theory see Zeilberger [25].

**Definition 2.5** (Dyck word and the Stanley order)**.** Set $w(\text{leaf}) = \varepsilon$ and $w\big((L,R)\big) = U\,w(R)\,D\,w(L)$. This is a Dyck word of length $2(n-1)$. Its height function $H_{w(T)}(k)$ is the number of $U$'s minus the number of $D$'s among the first $k$ letters. We write $T \le_{\mathrm{St}} U$ when $H_{w(T)}(k) \le H_{w(U)}(k)$ for all $k$.

With this convention the fully right-nested tree $(x_1,(x_2,(\dots)))$ has the highest path and the fully left-nested tree the lowest, which is the orientation used in [19]. The Stanley order [21] *extends* the Tamari order: $T \le_{\mathcal{T}} U$ implies $T \le_{\mathrm{St}} U$, and the two differ already at $n = 4$.

**Definition 2.6** (Right depth)**.** $r_i(T)$ is the number of right branches on the path from the root of $T$ to its $i$th leaf; left branches are not counted. Thus $r(\text{leaf}) = (0)$ and

$$r\big((L,R)\big) = \big(r(L),\, 1 + r(R)\big),$$

where $1 + r(R)$ means adding 1 to each entry.

**Lemma 2.7.** *For every $T \in \mathcal{T}_n$ we have $r_1(T) = 0$, and if $n \ge 2$ then $r_2(T) = 1$. Consequently two trees with the same number of leaves can differ in right depth only at an index $i \ge 3$.*

*Proof.* Induction on $T$. A leaf gives $r = (0)$. For $T = (L,R)$ the first entry of $r(T)$ is the first entry of $r(L)$, which is 0 by induction. For the second entry there are two cases: if $L$ is a leaf then $r(T) = (0,\, 1 + r(R))$ and the second entry is $1 + r_1(R) = 1$; if $L$ has at least two leaves the second entry of $r(T)$ is $r_2(L)$, which is 1 by induction. □

### 2.4 The semi-associative law, and one lemma used throughout

The inequality

$$\mathrm{H}_r\big(\mathrm{H}_r(x,y),z\big) \;\le\; \mathrm{H}_r\big(x,\mathrm{H}_r(y,z)\big) \tag{$\mathrm{HT}_r$}$$

is exactly the assertion that a right rotation, applied with $x, y, z$ the values of the three subtrees, does not decrease the value.

The following single lemma is used in both parts of the paper: in Part II it is what makes ($\mathrm{HT}_r$) true at rank 3 and, through the induction of Section 12, at every rank; in Part I it is one of the two absorption steps of Section 5.

**Lemma 2.8.** *If $y \ge 2$ then $yz \le y^z$ for every $z$. If moreover $z \ge 3$ the inequality is strict.*

*Proof.* The cases $z = 0, 1$ are immediate. For $z \ge 1$ the induction step is $y(z+1) \le 2yz \le y\,y^z = y^{z+1}$, using $z + 1 \le 2z$ and, for the middle inequality, $2z \le yz \le y^z$. For strictness, $3y < y^3$ because $y^2 \ge 4 > 3$; and if $yz < y^z$ with $z \ge 3$, then $y(z+1) \le 2yz < 2y^z \le y^{z+1}$. □

**Proposition 2.9.** *If $y \ge 2$ then $yz = y^z$ if and only if $z = 1$ or $(y,z) = (2,2)$.*

*Proof.* If $z = 0$ then $yz = 0 \ne 1 = y^z$. If $z = 1$ both sides are $y$. If $z = 2$ then $yz = y^z$ reads $2y = y^2$, i.e. $y = 2$ after cancelling $y > 0$. If $z \ge 3$ the inequality is strict by Lemma 2.8. The converse is immediate. □

# Part I. Rank three: the evaluation order

## 3 Rank three: the semi-associative law

At rank 3 the collapsing law $(x^y)^z = x^{yz}$ reduces ($\mathrm{HT}_r$) to a comparison of exponents.

*Proof of Theorem 1.4.* Since $(x^y)^z = x^{yz}$, the inequality follows from Lemma 2.8 and monotonicity of $t \mapsto x^t$ for $x \geq 1$. For $x \geq 2$ the map $t \mapsto x^t$ is injective, so equality holds iff $yz = y^z$, and Proposition 2.9 applies. For the last claim, $(x^1)^z = x^z$ while $x^{(1^z)} = x$, and $x^z > x$ whenever $x, z \geq 2$. □

**Example 3.1** (Failure of strictness)**.** Take $a = 2$ and $n = 4$. The five bracketings evaluate under $\mathrm{H}_3$ to

$$((aa)a)a = (a(aa))a = (aa)(aa) = 256, \qquad a((aa)a) = a(a(aa)) = 65536.$$

In particular the Tamari cover $((aa)a)a \lessdot (aa)(aa)$ has equal endpoints, so $\mathrm{ev}_{3,2}$ is monotone but not strictly monotone; it is also not injective.

The coincidence $(a(aa))a = (aa)(aa)$ is not special to $a = 2$: for every $a$ both sides equal $a^{a^{a+1}}$. Göbel and Nederpelt [6] already use this pair to show that different bracketings can have the same value when every leaf carries the same real number $x > 1$. Their standard form turns a bracketing with $n$ leaves into a rooted tree with $n - 1$ edges, and bracketings with the same tree have the same value; hence $\mathrm{ev}_{3,a}$ takes at most as many values on $\mathcal{T}_n$ as there are rooted trees with $n - 1$ edges [6, Theorem 2.2], fewer than $C_{n-1}$ once $n \geq 4$. Guy and Selfridge [10] note the same identity in their study of how many distinct values the bracketings of $a \star a \star \cdots \star a$ take under exponentiation, and conclude that "exponentiation is not completely non-associative".

# 4 The Stanley order in right-depth coordinates

**Proposition 4.1** (Coordinate presentation; known)**.** *For $T, U \in \mathcal{T}_n$,*

$$T \leq_{\mathrm{St}} U \quad \Longleftrightarrow \quad r_i(T) \leq r_i(U) \ \text{ for all } i = 1, \ldots, n.$$

*Proof.* Let $q(\text{leaf}) = \varepsilon$ and $q\big((L, R)\big) = q(L)\, U\, q(R)\, D$; this is the word obtained from postfix notation by writing $U$ for a leaf and $D$ for an operator and deleting the initial $U$. An induction on the tree shows that $q(T)$ is a Dyck word and that the height immediately after its $j$th up-step equals $r_{j+1}(T)$: in $q(L)\, U\, q(R)\, D$ the up-steps of $q(L)$ keep their heights, the displayed $U$ has height 1, and the up-steps of $q(R)$ have their heights raised by 1, which matches Definition 2.6.

Let $p_j(T)$ be the position of the $j$th up-step, positions counted from 1. Among the first $p_j$ letters there are $j$ up-steps and $p_j - j$ down-steps, so $r_{j+1}(T) = 2j - p_j(T)$. Since the height at position $k$ is $2u(k) - k$ with $u(k)$ the number of up-steps so far, one Dyck word lies weakly below another exactly when each of its up-steps occurs no earlier, i.e. exactly when $p_j$ is weakly larger for every $j$, i.e. exactly when $r_{j+1}$ is weakly smaller for every $j$.

Finally, write $\bar{u}$ for the word obtained from $u$ by reversing it and exchanging $U$ with $D$. Then $\overline{q(T)} = w(T)$: indeed $\overline{q(L)\, U\, q(R)\, D} = U\, \overline{q(R)}\, D\, \overline{q(L)}$, which is $U\, w(R)\, D\, w(L) = w\big((L, R)\big)$ by induction. For a word $u$ of length $N$ with equally many $U$'s and $D$'s one has $H_{\bar{u}}(k) = H_u(N - k)$, since the first $k$ letters of $\bar{u}$ are the last $k$ letters of $u$ with the two symbols exchanged. Hence $q(T)$ lies weakly below $q(U)$ if and only if $w(T)$ does below $w(U)$, and the comparison may be read off in either convention. □

**Remark 4.2.** As said in the introduction, Proposition 4.1 is known. Knuth's exercise 28 defines the Stanley lattice on $n$-node forests by comparing depth coordinates, and his Fig. 41 lists the 14 forests of order 4 by those coordinates [14]. The passage between his indexing and ours is the shift $c_j = r_{j+1} - 1$, which comes from deleting the root: a tree with $n$ leaves gives a forest with $n - 1$ nodes. Under it the five vectors of Example 4.3 become the order-3

sequences 012, 011, 001, 010, 000, and the fourteen vectors for $n = 5$ reproduce, as a set, exactly the fourteen sequences printed in Fig. 41. That right-depth sequences determine the bracketing is 2.8 of [4], where the sequences occurring are also characterised (in their notation, with $d_i = r_i$, they are the $(d_1, \ldots, d_n)$ with $d_1 = 0$, $d_2 = 1$ and $1 \le d_{i+1} \le d_i + 1$); the identification with DFS depth is [17, §2.5].

**Example 4.3.** For $n = 4$ the five trees have right-depth vectors

| | | | |
|---|---|---|---|
| $(a(b(cd)))$ | $(0,1,2,3)$ | $((a(bc))d)$ | $(0,1,2,1)$ |
| $(a((bc)d))$ | $(0,1,2,2)$ | $(((ab)c)d)$ | $(0,1,1,1)$ |
| $((ab)(cd))$ | $(0,1,1,2)$ | | |

Exactly one pair is incomparable, namely $((ab)(cd))$ and $((a(bc))d)$: their vectors $(0,1,1,2)$ and $(0,1,2,1)$ differ in both directions. This is the pair called $C$ and $D$ in [19].

## 4.1 The covering relation

The next two lemmas describe the covers of the Stanley order. They are used twice: in Section 7 for rank three and in Section 14 for every rank above it.

**Lemma 4.4** (Cover description)**.** *$T \lessdot U$ in the Stanley order if and only if $w(U)$ is obtained from $w(T)$ by replacing one occurrence of $DU$ by $UD$.*

This is the answer to Knuth's exercise 28(c) [14], where the same operation is described on nested parenthesis strings as "changing `)(` to `()`"; in the discussion of [19] it is also stated, in terms of peaks and valleys by Rubey and in reverse Polish form by Chow. We use it as known and claim no priority for it. For completeness, and so that the reader need not consult the artifact, here is a direct proof; a machine-checked proof is also included (Section 16), because the formalisation of Section 14 rests on it.

*Proof of Lemma 4.4.* Replacing an occurrence of $DU$ by $UD$ raises exactly one height, by 2, and leaves the word balanced and nonnegative. No word lies strictly between the two: they agree at every other position, and at the position in question a height must have the parity of its index, so a value strictly between $h - 1$ and $h + 1$ is impossible.

Conversely let $w < v$ be Dyck words of the same length, and let $k$ be the first position at which they differ. The heights agree before $k$, so $w$ has $D$ and $v$ has $U$ there, and at position $k$ the height of $v$ exceeds that of $w$ by exactly 2. Let $\ell > k$ be the first position after $k$ at which $w$ has $U$; such a position exists, because the two words have equally many up-steps and $v$ has used one more of them by position $k$. Between $k$ and $\ell - 1$ the word $w$ takes only down-steps, while each letter of $v$ changes its height by $\pm 1$; so on that range the excess of the height of $v$ over that of $w$ never decreases, and it is at least 2 throughout. Swapping the $DU$ that $w$ carries at positions $\ell - 1, \ell$ therefore produces a Dyck word $w'$ with $w < w' \le v$, and if $v$ covers $w$ then $w' = v$. The first-return decomposition identifies Dyck words with trees, so the statement transfers to the Stanley order on trees. □

**Lemma 4.5** (Local form of a cover)**.** *Every tree $R$ factors uniquely along its leftmost leaf as*

$$R = L_{\mathbf{D}}(c) := \big(\cdots((c \cdot D_1) \cdot D_2)\cdots D_m\big), \qquad m \ge 0,$$

*where $c$ is that leaf. A pair $T \lessdot U$ of the Stanley order is exactly a pair*

$$T = \big((A \cdot B) \cdot L_{\mathbf{D}}(c)\big) \;\lessdot\; U = \big(A \cdot L_{\mathbf{D}}\big((B \cdot c)\big)\big)$$

*occurring inside a common tree context.*

*Proof.* The factorisation is obtained by peeling off left children until the leftmost leaf is reached, and is clearly unique. Write

$$V = U\,w(D_m)\,D \cdots U\,w(D_1)\,D.$$

Since $w$ of a leaf is empty, the recursion $w\big((L,R)\big) = U\,w(R)\,D\,w(L)$ gives $w\big(L_{\mathbf{D}}(c)\big) = V$, and hence

$$w(T) = U\,V\,\mathbf{D}\,\mathbf{U}\,w(B)\,D\,w(A), \quad w(U) = U\,V\,\mathbf{U}\,\mathbf{D}\,w(B)\,D\,w(A),$$

the second because $w\big(L_{\mathbf{D}}((B \cdot c))\big) = V\,U\,D\,w(B)$. So the displayed pair replaces exactly one $DU$ by $UD$.

Conversely, let a $DU$ be given in $w(T) = U\,w(R)\,D\,w(L)$. It lies inside $w(R)$, inside $w(L)$, or consists of the $D$ following $w(R)$ together with the first letter of $w(L)$; the last happens exactly when $w(L)$ is non-empty, that is when $L = (A \cdot B)$, and then $T$ has the displayed form with $R = L_{\mathbf{D}}(c)$. The first two cases are the same statement for a subtree, so induction on $T$ finishes the proof. □

# 5 Absorbing a power into a tower

**Definition 5.1.** $F_0(t) = t$ and $F_{h+1}(t) = 2^{F_h(t)}$. All arguments below are integers $\geq 2$, and each $F_h$ is strictly increasing on them. Two facts are used repeatedly and follow by induction on $h$: $F_h(t) \leq F_{h+1}(t)$, because $z \leq 2^z$; and

$$F_{h+1}(t) = F_h\big(2^t\big),$$

so that replacing the argument $t$ by $2^t$ and adding one to the height are the same operation.

**Lemma 5.2.** *For every integer $c \geq 1$ we have $c \leq 2^{c-1}$.*

*Proof.* Induction on $c$: it holds at $c = 1$, and $c + 1 \leq 2c \leq 2 \cdot 2^{c-1}$. □

Together with Lemma 2.8, which gives $cy \leq y^c$ for $c \geq 1$ and $y \geq 2$, this is all the elementary arithmetic the present section needs. Note that Lemma 2.8 is the same inequality that drives the whole hierarchy in Part II; it is used here in the form $c\,y \leq y^c$ and there in the form $yz \leq y^z$.

**Lemma 5.3** (Absorption)**.** *For all $h \geq 0$, integers $z \geq 2$ and $c \geq 1$,*

$$F_h(z)^{\,c} \;\leq\; F_h(z^c).$$

*Proof.* Induction on $h$. At $h = 0$ both sides equal $z^c$. Assuming the claim at $h$,

$$F_{h+1}(z)^c = 2^{\,c\,F_h(z)} \leq 2^{\,F_h(z)^c} \leq 2^{\,F_h(z^c)} = F_{h+1}(z^c),$$

the first inequality by Lemma 2.8 with $y = F_h(z) \geq 2$ and the second by the induction hypothesis. □

**Lemma 5.4** (Absorbing the base)**.** *For integers $c, y \geq 2$ we have $c^{\,y} \leq 2^{\,y^c}$.*

*Proof.* $c^y \leq (2^c)^y = 2^{cy} \leq 2^{y^c}$ by Lemma 2.8. □

## 6 Bounds when a single leaf is enlarged

**Definition 6.1.** For a tree $T$ and a leaf index $i$, walk from that leaf to the root. At each step the path leaves a sibling subtree behind; let $c_1, \dots, c_m$ be the values of those sibling subtrees when *all* their leaves are set to 2, and put

$$K_i(T) = \prod_{\nu=1}^{m} c_\nu,$$

the empty product being 1. Each $c_\nu$ is an integer $\geq 2$ and does not depend on the value placed at leaf $i$.

**Definition 6.2.** $v_{T,i}(t) = \mathrm{ev}_3(T)(2, \dots, 2, t, 2, \dots, 2)$, the value of $T$ when leaf $i$ carries $t$ and every other leaf carries 2.

**Proposition 6.3** (Bounds)**.** *For every integer $t \geq 2$,*

$$F_{r_i(T)}(t) \;\leq\; v_{T,i}(t) \;\leq\; F_{r_i(T)}\big(t^{K_i(T)}\big).$$

*Proof.* Induction along the path from leaf $i$ to the root. At the leaf the value is $t$, the right depth is 0 and the product is 1, so both bounds are equalities.

Suppose the current subtree has value $v$, right depth $h$ so far and sibling product $K$ so far, and that $F_h(t) \leq v \leq F_h(t^K)$. Let $c \geq 2$ be the value of the next sibling subtree, all of its leaves being 2.

If the subtree containing leaf $i$ is the *left* child, the new value is $v^c$, the right depth is unchanged and the product becomes $Kc$. Then $v^c \geq v \geq F_h(t)$, and by Lemma 5.3,

$$v^c \leq F_h(t^K)^c \leq F_h\big((t^K)^c\big) = F_h(t^{Kc}).$$

If it is the *right* child, the new value is $c^v$, the right depth becomes $h+1$ and the product becomes $Kc$. Then $c^v \geq 2^v \geq 2^{F_h(t)} = F_{h+1}(t)$, and by Lemmas 5.4 and 5.3,

$$c^v \leq 2^{v^c} \leq 2^{F_h(t^K)^c} \leq 2^{F_h(t^{Kc})} = F_{h+1}(t^{Kc}).$$

Either way the hypothesis is restored, and at the root it is the assertion. □

**Remark 6.4.** The qualitative content of Proposition 6.3 — that $r_i(T)$ is the level at which leaf $i$ sits in the tower — is the observation of 3.4 in [4]. What the proposition adds is the two-sided estimate, and in particular the upper bound, in which the whole effect of the other leaves is compressed into the single factor $K_i(T)$ in the exponent of the argument.

## 7 The Stanley order at rank three

The inclusion $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ at rank three is Chow's, proved in [19] by an argument on reverse Polish notation. With Lemmas 4.4 and 4.5 in hand it has a short direct proof, which we give because it makes Theorem 1.1 self-contained and because the same computation locates the equality cases. **The priority is Chow's**; what follows is an alternative proof, not a new result.

**Theorem 7.1** (Stanley monotonicity at rank three)**.** *Let $T, U \in \mathcal{T}_n$ carry the same leaf labels, all $\geq 2$. If $T \leq_{\mathrm{St}} U$ then $\mathrm{ev}_3(T) \leq \mathrm{ev}_3(U)$. Consequently $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$.*

*Proof.* By Lemma 4.4 it suffices to treat one cover, and by Lemma 4.5 we may take

$$T = \big((A \cdot B) \cdot L_{\mathbf{D}}(c)\big), \qquad U = \big(A \cdot L_{\mathbf{D}}\big((B \cdot c)\big)\big)$$

inside a common context. Write $a, b$ for the values of $A, B$, let $c$ also denote the value of the leftmost leaf of $L_{\mathbf{D}}(c)$, and let $d_1, \dots, d_m$ be the values of $D_1, \dots, D_m$; all are $\geq 2$. At rank three the collapsing law $(x^y)^z = x^{yz}$ gives

$$G(t) := \big(\cdots((t^{d_1})^{d_2})\cdots\big)^{d_m} = t^K, \qquad K := d_1 d_2 \cdots d_m \ \geq 1,$$

the empty product being 1. Hence

$$\mathrm{ev}_3(T) = (a^b)^{G(c)} = a^{\,b\,c^K}, \qquad \mathrm{ev}_3(U) = a^{\,G(b^c)} = a^{\,(b^c)^K} = a^{\,b^{Kc}},$$

and since $a \geq 2$ it is enough to compare the exponents. By Lemma 5.2, $c \leq 2^{c-1} \leq b^{c-1}$, so $c^K \leq b^{K(c-1)}$ and therefore

$$b\,c^K \ \leq \ b^{\,K(c-1)+1} \ \leq \ b^{\,Kc}, \tag{1}$$

the last step because $K \geq 1$. So $\mathrm{ev}_3(T) \leq \mathrm{ev}_3(U)$, and since $\mathrm{H}_3$ is increasing in each argument on values $\geq 2$ the inequality propagates through the surrounding context. Chaining covers gives the theorem. □

**Corollary 7.2** (Equality along a Stanley cover at rank three)**.** *In the situation above,* $\mathrm{ev}_3(T) = \mathrm{ev}_3(U)$ *if and only if* $m = 0$ *and* $b = c = 2$*; that is, if and only if* $L_{\mathbf{D}}(c)$ *is a leaf and both it and* $B$ *are leaves labelled* 2*.*

*Proof.* The second inequality of (1) is strict unless $K(c-1)+1 = Kc$, that is unless $K = 1$; and $K = 1$ forces $m = 0$, since every $d_j$ is at least 2. When $K = 1$ the first inequality reads $bc \leq b^c$, which by Proposition 2.9 is an equality exactly when $c = 1$ or $(b, c) = (2, 2)$; the first is excluded because $c \geq 2$. □

**Remark 7.3.** When $m = 0$ the cover of Lemma 4.5 reads $((A \cdot B) \cdot c) \to (A \cdot (B \cdot c))$, which is a Tamari rotation. So Corollary 7.2 contains Corollary 13.2 at rank three, and says in addition that a Stanley cover which is *not* a Tamari rotation — one with $m \geq 1$ — is always strict.

## 8 Separation, and the main theorem

**Lemma 8.1** (Threshold)**.** *For every integer* $K \geq 1$*, putting* $M = 2K^2$ *we have* $2^M > M^K$*.*

*Proof.* By Lemma 5.2, $K \leq 2^{K-1}$, so $M^K = (2K^2)^K \leq \big(2 \cdot 2^{2(K-1)}\big)^K = 2^{(2K-1)K} < 2^{2K^2} = 2^M$. □

*Proof of Theorem 1.2.* Write $s = r_i(U)$, $K = K_i(U)$ and $M = 2K^2$, and give leaf $i$ the value $M$ with every other leaf equal to 2. By hypothesis $r_i(T) \geq s + 1$. Using Proposition 6.3 on both sides, the monotonicity of the $F_h$ in $h$ for arguments $\geq 2$, the identity $F_{s+1}(M) = F_s(2^M)$, the strict monotonicity of $F_s$ and Lemma 8.1,

$$v_{T,i}(M) \ \geq \ F_{r_i(T)}(M) \ \geq \ F_{s+1}(M) \ = \ F_s(2^M) \ > \ F_s(M^K) \ \geq \ v_{U,i}(M). \qquad \square$$

*Proof of Theorem 1.1.* The implication $T \leq_{\mathrm{St}} U \Rightarrow T \leq_{\mathrm{ev}} U$ is Theorem 7.1. For the converse, suppose $T \not\leq_{\mathrm{St}} U$. By Proposition 4.1 there is an index $i$ with $r_i(T) > r_i(U)$, and Theorem 1.2 produces an assignment at which $\mathrm{ev}_3(T) > \mathrm{ev}_3(U)$; hence $T \not\leq_{\mathrm{ev}} U$. □

**Corollary 8.2.** *For every $n$ the $C_{n-1}$ trees in $\mathcal{T}_n$ induce pairwise distinct functions on $\mathbb{N}_{\geq 2}^n$.*

*Proof.* If $T \neq U$ then $r(T) \neq r(U)$, by Proposition 4.1 and the antisymmetry of $\leq_{\mathrm{St}}$ (or directly by 2.8 of [4]), so some coordinate differs and Theorem 1.2 separates the two values. □

**Remark 8.3.** Corollary 8.2 says that the associative spectrum of exponentiation on $\mathbb{N}_{\geq 2}$ is the Catalan number $C_{n-1}$, which is 3.4 of [4]; that proof substitutes distinct primes and appeals to unique factorisation. The present argument recovers it and says in addition which of the two values is the larger.

**Remark 8.4.** Applying Theorem 1.2 to the ordered pairs $(T, U)$ and $(U, T)$ in turn shows that two Stanley-incomparable trees are separated in *both* directions, by two assignments each of which differs from $(2, \ldots, 2)$ in a single coordinate. This is the statement Chow left open.

# 9 A uniform finite test family

**Lemma 9.1.** *For all $p, q \geq 0$,*

$$\textit{(i)}\ F_p(2)\,F_q(2) \leq F_{p+q+1}(2), \qquad \textit{(ii)}\ F_p(2)^{\,F_q(2)} \leq F_{p+q+1}(2).$$

*Proof.* Every $F_h(2)$ is one of $2, 4, 16, 65536, \ldots$, and $z^2 \leq 2^z$ holds at $z = 2$, at $z = 4$ and for every real $z \geq 4$; hence $F_h(2)^2 \leq 2^{F_h(2)} = F_{h+1}(2)$ for all $h$. For (i) we may assume $p \leq q$, and then $F_p(2)F_q(2) \leq F_q(2)^2 \leq F_{q+1}(2) \leq F_{p+q+1}(2)$. For (ii), if $p = 0$ then $F_0(2)^{F_q(2)} = 2^{F_q(2)} = F_{q+1}(2)$; and if $p \geq 1$ then $F_p(2) = 2^{F_{p-1}(2)}$, so by (i)

$$F_p(2)^{\,F_q(2)} = 2^{\,F_{p-1}(2)F_q(2)} \leq 2^{\,F_{p+q}(2)} = F_{p+q+1}(2).$$ □

**Lemma 9.2.** *A tree with $k$ leaves has value at most $F_{k-1}(2)$ when every leaf carries 2.*

*Proof.* Induction on the tree. A leaf gives $2 = F_0(2)$. A tree $(L, R)$ with $k_L$ and $k_R$ leaves gives $\mathrm{ev}_3(L)^{\mathrm{ev}_3(R)} \leq F_{k_L-1}(2)^{\,F_{k_R-1}(2)} \leq F_{k_L+k_R-1}(2)$, the last step by Lemma 9.1(ii). □

**Lemma 9.3.** *For every $n \geq 2$, every $T \in \mathcal{T}_n$ and every leaf $i$, $K_i(T) \leq F_{n-2}(2)$.*

*Proof.* The sibling subtrees met on the path from leaf $i$ to the root partition the remaining $n-1$ leaves; let $k_1, \ldots, k_m$ be their numbers of leaves, so $\sum_\nu k_\nu = n-1$. By Lemma 9.2, $c_\nu \leq F_{k_\nu-1}(2)$. Applying Lemma 9.1(i) $m-1$ times,

$$K_i(T) = \prod_{\nu=1}^{m} c_\nu \ \leq\ \prod_{\nu=1}^{m} F_{k_\nu-1}(2) \ \leq\ F_{(\sum_\nu k_\nu)-1}(2) = F_{n-2}(2),$$

since $F_{a-1}(2)F_{b-1}(2) \leq F_{a+b-1}(2)$ is Lemma 9.1(i) with $p = a-1$ and $q = b-1$. □

*Proof of Corollary 1.3.* One direction is Theorem 7.1. For the other, suppose the $n-2$ displayed inequalities hold but $T \not\leq_{\mathrm{St}} U$. By Lemma 2.7 any index at which the two right-depth vectors differ satisfies $i \geq 3$, so we may choose such an $i$ with $r_i(T) > r_i(U)$; write $s = r_i(U)$. Lemma 8.1 with $K = F_{n-2}(2)$ gives $2^{M_n} > M_n^K$, and $K_i(U) \leq K$ by Lemma 9.3, so $2^{M_n} > M_n^{K_i(U)}$. The chain in the proof of Theorem 1.2 now runs verbatim with $M_n$ in place of $M$:

$$\mathrm{ev}_3(T)(\vec{a}^{(i)}) \ \geq\ F_{s+1}(M_n) \ =\ F_s\big(2^{M_n}\big) \ >\ F_s\big(M_n^{K_i(U)}\big) \ \geq\ \mathrm{ev}_3(U)(\vec{a}^{(i)}),$$

contradicting the $i$th displayed inequality. □

**Remark 9.4.** We do not claim $M_n$ is the least threshold that works. For $n = 4$ it is $M_4 = 2 \cdot 16^2 = 512$; for $n = 5$ it is $2 \cdot 65536^2 = 8589934592$.

# Part II. Up the hierarchy: Tamari and Stanley orders

## 10 Rank four

At rank 4 we know of no identity that would play the role of $(x^y)^z = x^{yz}$, and a different argument is required. The proof below is by induction on $z$; the base case $z = 1$ must be treated separately, since the inductive hypothesis at $z = 0$ is too weak to survive the step.

**Lemma 10.1** (Squaring)**.** *If $x \geq 2$ and $m \geq 1$ then $(x \uparrow\uparrow m)^2 \leq x \uparrow\uparrow (m+1)$.*

*Proof.* Induction on $m$. For $m = 1$ the claim is $x^2 \leq x^x$, which holds as $x \geq 2$. For the step put $t = x \uparrow\uparrow m$; then $(x \uparrow\uparrow (m+1))^2 = (x^t)^2 = x^{2t} \leq x^{(x^t)} = x \uparrow\uparrow (m+2)$, where $2t \leq 2^t \leq x^t$ uses $t \geq 1$ and $x \geq 2$. □

**Remark 10.2.** The direct inequality $t^2 \leq x^t$ is *false* in general — take $x = 2$, $t = 3$, where $t^2 = 9 > 8 = x^t$ — so the inductive formulation in Lemma 10.1 is essential.

**Lemma 10.3.** *If $y \geq 2$ and $m \geq 2$ then $m + 2 \leq y^m$. If moreover $m \geq 3$ then $m + 3 \leq y^m$.*

*Proof.* It suffices to use $y^m \geq 2^m$. From $2^2 = 4 = 2 + 2$ the bound $2^m \geq m + 2$ propagates, because $2(m + 2) \geq (m + 1) + 2$; from $2^3 = 8 \geq 3 + 3$ the bound $2^m \geq m + 3$ propagates, because $2(m + 3) \geq (m + 1) + 3$. □

*Proof of Theorem 1.5.* Write $y = y' + 1$ with $y' \geq 1$ and induct on $z$. For $z = 0$ the left side is 1 and the right side is $x \geq 1$. For $z = 1$ both sides equal $x \uparrow\uparrow y$. Assume $z \geq 1$ and set $m := y \uparrow\uparrow z$, so $m \geq y \geq 2$ and $y' \leq m$. By Lemma 10.3 we have $y^m \geq m + 2$, so $y^m = k + 1$ with $k \geq m + 1$. Writing $A := x \uparrow\uparrow y = x^{(x\uparrow\uparrow y')}$ we compute

$$\begin{aligned}(x \uparrow\uparrow y) \uparrow\uparrow (z+1) = A^{(x\uparrow\uparrow y)\uparrow\uparrow z} \;\leq\; A^{x\uparrow\uparrow m} \;=\; \big(x^{x\uparrow\uparrow y'}\big)^{x\uparrow\uparrow m} \;=\; x^{(x\uparrow\uparrow y')(x\uparrow\uparrow m)} \\ \leq x^{(x\uparrow\uparrow m)^2} \leq x^{x\uparrow\uparrow(m+1)} \leq x^{x\uparrow\uparrow k} = x \uparrow\uparrow (k+1) = x \uparrow\uparrow \big(y \uparrow\uparrow (z+1)\big),\end{aligned}$$

using the inductive hypothesis, $x \uparrow\uparrow y' \leq x \uparrow\uparrow m$, Lemma 10.1, and $m + 1 \leq k$. This proves the inequality.

For strictness, note that the penultimate step is strict as soon as $m \geq 3$: then $y^m \geq m+3$ by Lemma 10.3, hence $m + 1 < k$ and $x \uparrow\uparrow (m + 1) < x \uparrow\uparrow k$. Now $m = y \uparrow\uparrow z \geq 3$ holds whenever $z \geq 2$ (then $m \geq 4$), and also when $z = 1$ and $y \geq 3$ (then $m = y$). This gives strictness at $z + 1$ for every $z + 1 \geq 2$ except $(y, z + 1) = (2, 2)$, which we compute directly:

$$(x \uparrow\uparrow 2) \uparrow\uparrow 2 = (x^x)^{x^x} = x^{x \cdot x^x} = x^{x^{x+1}}, \qquad x \uparrow\uparrow (2 \uparrow\uparrow 2) = x \uparrow\uparrow 4 = x^{x^{x^x}},$$

and $x + 1 < x^x$ for $x \geq 2$. Hence the inequality is strict for every $z \geq 2$. □

## 11 The dichotomy

*Proof of Corollary 1.8.* Order-preservation at every rank $r \geq 3$ is Proposition 1.6. Non-strictness at rank 3 is Example 3.1. Strictness at rank 4 is Theorem 1.5; for general $r \geq 4$ it

is Theorem 1.7, which is proved in Section 12 and rests on Theorem 1.10(3). (The present section proves only the rank-4 case, so that the dichotomy is available before the general induction.) □

*Proof of Proposition 1.6.* Induction on the derivation of the cover. At the root the claim is Theorem 1.9 instantiated at the values of the three subtrees, which are all $\geq 2$ because $a \geq 2$. If the rotation takes place inside the left subtree, one uses monotonicity of $\mathrm{H}_r(\cdot, v)$ in its first argument; inside the right subtree, monotonicity of $\mathrm{H}_r(u, \cdot)$ in its second argument, valid for $u \geq 2$. Both monotonicity statements hold for every $r \geq 1$ and are proved by a double induction on the rank and on the argument. The formula for the right comb is immediate from $\mathrm{H}_{r+1}(a, n+1) = \mathrm{H}_r(a, \mathrm{H}_{r+1}(a, n))$ together with $\mathrm{H}_{r+1}(a, 1) = a$. □

The mechanism is worth stating plainly. At rank 3 the identity $(x^y)^z = x^{yz}$ turns $(\mathrm{HT}_r)$ into $yz \leq y^z$, an inequality which is *tight*: it becomes an equality at $(y, z) = (2, 2)$. Each of the three coincidences in Example 3.1 is an instance of this one equality, applied at a node whose two lower subtrees both evaluate to 2: the cover $((aa)a)a \lessdot (aa)(aa)$ is the root rotation with $(x, y, z) = (4, 2, 2)$, and the pairs $((aa)a)a$, $(a(aa))a$ and $a((aa)a)$, $a(a(aa))$ differ only in a subtree $(aa)a = a(aa) = 16$, which is the root rotation with $(x, y, z) = (2, 2, 2)$. At rank 4 Theorem 1.5 shows that no equality with $z \geq 2$ occurs at all, and the gap is already visible at the smallest instance: for $x = y = z = 2$ the two sides are 256 and 65536.

## 12 General rank

We now prove Theorem 1.9. The engine is the following inequality; write

$$\mathrm{H}_{r-1}\big(\mathrm{H}_r(x, a), \mathrm{H}_r(x, b)\big) \;\leq\; \mathrm{H}_r(x,\, a + b). \tag{$\mathrm{SUM}_r$}$$

**Proposition 12.1.** *Let $r \geq 3$ and $x \geq 2$.*

1. *If $(\mathrm{HT}_{r-1})$ holds, then $(\mathrm{SUM}_r)$ holds for all $a, b \geq 1$.*
2. *If $(\mathrm{SUM}_r)$ holds and $u + v \leq \mathrm{H}_{r-1}(u, v)$ for all $u, v \geq 2$, then $(\mathrm{HT}_r)$ holds for all $y \geq 2$.*

*Proof.* (1) Induct on $a$. For $a = 1$ both sides are equal:

$$\mathrm{H}_{r-1}\big(\mathrm{H}_r(x, 1), \mathrm{H}_r(x, b)\big) = \mathrm{H}_{r-1}\big(x, \mathrm{H}_r(x, b)\big) = \mathrm{H}_r(x, b + 1).$$

For the step, $\mathrm{H}_r(x, a+1) = \mathrm{H}_{r-1}(x, \mathrm{H}_r(x, a))$, so by $(\mathrm{HT}_{r-1})$

$$\mathrm{H}_{r-1}\big(\mathrm{H}_{r-1}(x, \mathrm{H}_r(x, a)), \mathrm{H}_r(x, b)\big) \leq \mathrm{H}_{r-1}\big(x, \mathrm{H}_{r-1}(\mathrm{H}_r(x, a), \mathrm{H}_r(x, b))\big),$$

and the inductive hypothesis bounds the inner term by $\mathrm{H}_r(x, a+b)$, giving $\mathrm{H}_{r-1}(x, \mathrm{H}_r(x, a+b)) = \mathrm{H}_r(x, a+b+1)$.

(2) Induct on $z$, with $z = 0, 1$ as in Theorem 1.5. For the step set $m := \mathrm{H}_r(y, z) \geq 2$; then $\mathrm{H}_r(\mathrm{H}_r(x, y), z+1) = \mathrm{H}_{r-1}(\mathrm{H}_r(x, y), \mathrm{H}_r(\mathrm{H}_r(x, y), z)) \leq \mathrm{H}_{r-1}(\mathrm{H}_r(x, y), \mathrm{H}_r(x, m)) \leq \mathrm{H}_r(x, y + m) \leq \mathrm{H}_r(x, \mathrm{H}_{r-1}(y, m)) = \mathrm{H}_r(x, \mathrm{H}_r(y, z+1))$. □

*Proof of Theorem 1.9.* Induction on $r$. The base case $r = 3$ is Theorem 1.4. For the step, the inductive hypothesis is exactly $(\mathrm{HT}_{r-1})$, so Proposition 12.1(1) gives $(\mathrm{SUM}_r)$, and Proposition 12.1(2) then gives $(\mathrm{HT}_r)$, the hypothesis $u + v \leq \mathrm{H}_{r-1}(u, v)$ for $u, v \geq 2$ being Lemma 12.2. □

**Lemma 12.2.** *For $u, v \geq 2$ and $s \geq 2$ we have $u + v \leq \mathrm{H}_s(u, v)$.*

*Proof.* $u + v \leq uv = \mathrm{H}_2(u, v)$, and $\mathrm{H}_2(u, v) \leq \mathrm{H}_s(u, v)$ by monotonicity in the rank. □

## 12.1 Strictness

**Lemma 12.3.** *For $u, v \geq 2$ with $u \geq 3$ or $v \geq 3$, and $s \geq 2$, we have $u + v < \mathrm{H}_s(u, v)$. Equality $u + v = \mathrm{H}_s(u, v)$ occurs exactly at $u = v = 2$, where both sides are 4.*

*Proof.* $u + v < uv$ holds iff $(u - 1)(v - 1) > 1$, i.e. iff $u \geq 3$ or $v \geq 3$; and $uv = \mathrm{H}_2(u, v) \leq \mathrm{H}_s(u, v)$. At $u = v = 2$ we have $u + v = 4 = \mathrm{H}_2(2, 2) = \mathrm{H}_s(2, 2)$ by Remark 1.11. □

*Proof of Theorem 1.10.* (1) $\mathrm{H}_r(\mathrm{H}_r(x, y), 1) = \mathrm{H}_r(x, y) = \mathrm{H}_r(x, \mathrm{H}_r(y, 1))$ since $\mathrm{H}_r(a, 1) = a$. (2) is Theorem 1.4. For (3) we induct on $r \geq 4$, the base $r = 4$ being the strictness clause of Theorem 1.5. Let $r \geq 5$ and write $z = w + 1$ with $w \geq 1$, and $m := \mathrm{H}_r(y, w) \geq 2$.

If $y \geq 3$ or $w \geq 2$, then $y \geq 3$ or $m \geq 4$, so Lemma 12.3 makes the last step of the proof of Proposition 12.1(2) strict, and strictness propagates because $\mathrm{H}_r(x, \cdot)$ is strictly increasing.

The remaining case is $y = z = 2$. Put $A := \mathrm{H}_r(x, 2) = \mathrm{H}_{r-1}(x, x) \geq 4$. Then

$$\mathrm{H}_r(\mathrm{H}_r(x, 2), 2) = \mathrm{H}_{r-1}(A, A), \qquad \mathrm{H}_r(x, \mathrm{H}_r(2, 2)) = \mathrm{H}_r(x, 4) = \mathrm{H}_{r-1}\big(x, \mathrm{H}_{r-1}(x, A)\big),$$

using $\mathrm{H}_r(2, 2) = 4$ from Remark 1.11. Since $A = \mathrm{H}_{r-1}(x, x)$, the required inequality

$$\mathrm{H}_{r-1}\big(\mathrm{H}_{r-1}(x, x), A\big) < \mathrm{H}_{r-1}\big(x, \mathrm{H}_{r-1}(x, A)\big)$$

is the inductive hypothesis at rank $r - 1$ applied to $(x, x, A)$, and $A \geq 2$. □

We can now upgrade Proposition 1.6 to its strict form.

*Proof of Theorem 1.7.* Induction on the derivation of the cover, exactly as for Proposition 1.6 but with every inequality replaced by its strict counterpart. At the root the claim is Theorem 1.10(3) instantiated at the values of the three subtrees, which are all $\geq 2$ because $a \geq 2$ (so in particular the hypothesis $z \geq 2$ of that item is met). If the rotation takes place inside the left subtree, one needs

$$2 \leq u < u' \implies \mathrm{H}_r(u, v) < \mathrm{H}_r(u', v) \qquad (v \geq 1),$$

the strict form of monotonicity in the first argument; if it takes place inside the right subtree, one needs

$$1 \leq v < v' \implies \mathrm{H}_r(u, v) < \mathrm{H}_r(u, v') \qquad (u \geq 2,\ r \geq 2),$$

the strict form of monotonicity in the second argument. Both are proved by the same double induction on the rank and on the argument as their non-strict counterparts, and both apply here because every subtree value is $\geq 2$. □

The auxiliary facts used above — monotonicity of $\mathrm{H}_r$ in each argument, monotonicity in the rank, and $n + 1 \leq \mathrm{H}_r(x, n)$ for $x \geq 2$, $r \geq 2$, $n \geq 1$ — are proved from the recursion in Appendix B. The corresponding inductions are also part of the formalisation.

The hypothesis $y \geq 2$ of Theorem 1.9 is sharp, in the following precise sense: for every $r \geq 3$ and all $x, z \geq 2$, taking $y = 1$ reverses the inequality. Indeed $\mathrm{H}_r(x, 1) = x$, so the left side is $\mathrm{H}_r(x, z)$, whereas $\mathrm{H}_r(1, z) = 1$ makes the right side $\mathrm{H}_r(x, 1) = x$, and $\mathrm{H}_r(x, z) > x$ because $z \geq 2$. The restriction to $x, z \geq 2$ is necessary and not an artefact: for $x = 1$ both sides equal 1, since $\mathrm{H}_r(1, \cdot) = 1$; for $z = 1$ they are equal by Theorem 1.10(1); and for $z = 0$ the left side is 1 and the right side is $x$, so the original inequality still holds. For $x = 0$ the operations degenerate.

## 13 Arbitrary leaf values

Nothing in Part II so far has used the fact that the leaves all carry the same value. The local law ($\mathrm{HT}_r$), and its strict form, are statements about three independent arguments; only the evaluation $\mathrm{ev}_{r,a}$ was tied to a single label. Removing that restriction costs nothing and gives a strictly stronger statement, which is moreover the one that the discussion recalled in Section 17 was about.

**Theorem 13.1** (Arbitrary leaf values)**.** *Let $T$ be a labelled tree all of whose leaf labels are $\geq 2$, and let $T \lessdot U$ be a Tamari cover. Then*

$$\mathrm{ev}_r(T) \leq \mathrm{ev}_r(U) \quad (r \geq 3), \qquad \mathrm{ev}_r(T) < \mathrm{ev}_r(U) \quad (r \geq 4),$$

*and the same holds along the Tamari order itself, that is along the reflexive transitive closure of the covering relation in the first case and its transitive closure in the second.*

*Proof.* Rotation does not change the multiset of leaf labels, so the hypothesis is preserved along a cover, and every subtree of $T$ evaluates to at least 2 at every rank $r \geq 2$ by induction on the tree. The induction on the derivation of the cover is then the one already used for Proposition 1.6 and Theorem 1.7: at the root the claim is Theorem 1.9 respectively Theorem 1.10(3), instantiated at the three subtree values, which are independent of one another and all $\geq 2$; inside the left subtree one uses monotonicity, respectively strict monotonicity, of $\mathrm{H}_r(\cdot, v)$ in its first argument, and inside the right subtree the corresponding statement for $\mathrm{H}_r(u, \cdot)$. The passage to the closures is by induction on the length of the chain. □

We can now say exactly when a cover fails to be strict. At rank 3 the only non-trivial equality of ($\mathrm{HT}_r$) is $(y, z) = (2, 2)$, and in a labelled tree whose labels are all $\geq 2$ an *internal* node evaluates to at least $\mathrm{H}_r(2, 2) = 4$; so a subtree evaluates to 2 exactly when it is a leaf labelled 2.

**Corollary 13.2** (Equality along a cover)**.** *Let all labels be $\geq 2$ and consider the rotation $((A \cdot B) \cdot C) \to (A \cdot (B \cdot C))$ at rank 3. Its two sides are equal if and only if $B$ and $C$ are both leaves labelled 2. Consequently, for a single repeated label $a$, every cover is strict unless $(r, a) = (3, 2)$; and for $(r, a) = (3, 2)$ equality does occur, as Example 3.1 shows.*

*Proof.* By Theorem 1.4 the two sides agree exactly when $\mathrm{ev}_3(C) = 1$ or $(\mathrm{ev}_3(B), \mathrm{ev}_3(C)) = (2, 2)$. The first is impossible because every label is $\geq 2$. For the second, a subtree evaluates to 2 only if it is a leaf labelled 2, by the remark above. In a surrounding context each ancestor applies a strictly increasing function of the changed subtree value (Appendix B), so equality of the two whole trees is equivalent to equality at the rotation. For a single label $a$: if $a \geq 3$ no leaf is labelled 2, so no cover at rank 3 is an equality, and from rank 4 on Theorem 1.7 applies. □

Thus the collapse at rank 3 is not merely rare but completely located: it happens at exactly one local configuration, two leaves labelled 2 being rotated past each other, and from rank 4 on it does not happen at all.

Setting every label equal to $a$ recovers Proposition 1.6 and Theorem 1.7; the formalisation carries out this specialisation explicitly, so the earlier statements are corollaries rather than separate results.

Theorem 13.1 is what extends the rank-3 discussion of Section 17 to the whole hierarchy. At rank 3 it is subsumed by Theorem 1.1, since Tamari comparability implies Stanley comparability (Remark 1.12); what it contributes is the ranks $r \geq 4$, where the separation argument of Part I has no known analogue.

## 14 The Stanley order at every rank

Section 13 shows that evaluation respects the *Tamari* order at every rank. In fact more is true: it respects the *Stanley* order, which is strictly larger for $n \geq 4$. The proof needs the description of the covering relation, the *non-strict* nested upper bound stated by Leonardis, d'Atri and Caldarola [15, Lemma 4.8],

$$\mathrm{H}_r\big(\mathrm{H}_r(x,y),z\big) \;\leq\; \mathrm{H}_r(x,y+z) \qquad (r\geq 4,\; x\geq 2,\; y,z\geq 1), \tag{2}$$

and Theorem 1.10(3), which is proved in Section 12. The base $x = 2$ is needed, because a leaf may carry the label 2, and the published proof of [15, Lemma 4.8] does not treat it; Lemma C.1 in Appendix C therefore proves (2), in strict form, for every base $x \geq 2$. The proof below does *not* need the strict form; Remark 14.4 says what that stronger form buys.

The strictness comes from a gap that the hierarchy opens in its first argument.

**Lemma 14.1** (A two-unit gap in the base)**.** *For $r \geq 3$ and $t, d \geq 2$ we have $\mathrm{H}_r(t+1,d) \geq \mathrm{H}_r(t,d)+2$.*

*Proof.* Put $u = \mathrm{H}_r(t,d-1)$ and $v = \mathrm{H}_r(t+1,d-1)$. Since $d-1 \geq 1$ we have $v > u \geq 2$, both by monotonicity in the base and by $\mathrm{H}_r(t,1) = t$. As $r-1 \geq 2$, the two comparisons

$$\mathrm{H}_{r-1}(t,u) \;<\; \mathrm{H}_{r-1}(t,v) \;<\; \mathrm{H}_{r-1}(t+1,v)$$

are strict, the first by monotonicity in the second argument and the second by monotonicity in the first. Their outer terms are $\mathrm{H}_r(t,d)$ and $\mathrm{H}_r(t+1,d)$. Two strict inequalities between integers force a difference of at least 2. □

**Lemma 14.2** (The left spine widens the gap)**.** *Let $r \geq 3$, let $d_1,\dots,d_m$ be integers $\geq 2$ with $m \geq 1$, and put $G(t) = \big(\cdots((t * d_1) * d_2)\cdots * d_m\big)$ with $x * y = \mathrm{H}_r(x,y)$. Then $G(t+1) \geq G(t)+2$ for every $t \geq 2$, and hence $G(t+b) \geq G(t)+2b$ for every $b \geq 0$.*

*Proof.* The innermost map $t \mapsto t * d_1$ increases by at least 2 when $t$ does by 1, by Lemma 14.1. If $f$ is a strictly increasing map of the integers $\geq 2$ and $A \geq B+2$, then $f(B) < f(B+1) < f(B+2) \leq f(A)$, so $f(A) \geq f(B)+2$; each further map $t \mapsto t * d_j$ is such an $f$, so the gap survives the composition. Telescoping gives the second statement. □

**Theorem 14.3** (Stanley monotonicity at every rank)**.** *Let $r \geq 4$ and let $T, U \in \mathcal{T}_n$ be labelled trees with the same labels, all $\geq 2$. If $T <_{\mathrm{St}} U$ then $\mathrm{ev}_r(T) < \mathrm{ev}_r(U)$. Consequently $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ at every rank $r \geq 4$.*

*Proof.* By Lemma 4.4 it suffices to treat a single cover, and by Lemma 4.5 we may take $T = ((A \cdot B) \cdot L_{\mathbf{D}}(c))$ and $U = (A \cdot L_{\mathbf{D}}((B \cdot c)))$ inside a context. Write $x * y = \mathrm{H}_r(x,y)$, let $a, b$ be the values of $A, B$, let $c$ also denote the value of the leftmost leaf, let $d_j$ be the value of $D_j$, and set

$$G(t) = \big(\cdots((t * d_1) * d_2)\cdots * d_m\big), \qquad G(t) = t \text{ if } m = 0.$$

All of $a, b, c, d_j$ are $\geq 2$, and $\mathrm{ev}_r(T) = (a * b) * G(c)$, $\mathrm{ev}_r(U) = a * G(b * c)$.

If $m = 0$ then $G$ is the identity and the two values are $(a * b) * c$ and $a * (b * c)$, so the assertion is exactly the strict form of the semi-associative law, Theorem 1.10(3), applied at $(a, b, c)$ with $c \geq 2$.

Suppose $m \geq 1$. Since $\mathrm{H}_r$ dominates $\mathrm{H}_2$ on arguments $\geq 2$ we have $b * c \geq bc \geq b + c$, the second because $(b-1)(c-1) \geq 1$; and $G$ is strictly increasing, so Lemma 14.2 gives

$$G(b * c) \;\geq\; G(b+c) \;\geq\; G(c) + 2b \;>\; G(c) + b.$$

Now (2) bounds the left-hand value and strict monotonicity of $\mathrm{H}_r(a,\cdot)$ turns that last strict inequality into the conclusion:

$$\mathrm{ev}_r(T) = (a * b) * G(c) \;\le\; a * \big(b + G(c)\big) \;<\; a * G(b * c) = \mathrm{ev}_r(U).$$

Since $\mathrm{H}_r$ is strictly increasing in each argument on values $\ge 2$, the strict inequality propagates through the surrounding context, and chaining covers gives the theorem. □

**Remark 14.4.** The case $m = 0$ is where the argument has to be split, and the reason is visible at $b = c = 2$: there $b * c = 4 = b + c$, so $G(b * c) = G(b + c)$, and with $G$ the identity the chain above closes with an equality rather than a strict inequality. For $m \ge 1$ the gap of Lemma 14.2 supplies the missing unit; for $m = 0$ the cover is a Tamari rotation and Theorem 1.10(3) supplies it. With the *strict* form of (2) — Saibian's Theorem I [20], which reads $\mathrm{H}_r(\mathrm{H}_r(x,y),z) < \mathrm{H}_r(x,y+z)$ in the present notation and is proved here as Lemma C.1 — the split is unnecessary, since the first inequality above is then already strict. We print the route through the non-strict bound because it shows how little the Stanley covering relation needs; the Lean development follows the shorter route (Section 16).

**Remark 14.5.** Theorem 14.3 contains Theorem 13.1 for $r \ge 4$, because the Tamari order is contained in the Stanley order. It also removes one possibility from Section 18: at rank $r \ge 4$ the evaluation order cannot lie strictly between the Tamari and the Stanley orders. What remains open there is the reverse inclusion $\le_{\mathrm{ev}} \subseteq \le_{\mathrm{St}}$, for which the argument of Part I is unavailable: that argument rests on the collapsing law $(x^y)^z = x^{yz}$ through Lemma 5.3, and no analogue is known for tetration.

## 15 Computational verification

The rank-three results of Part I were checked by exact integer computation before and after they were proved. Trees were enumerated, right-depth vectors and Dyck words computed independently, and the two descriptions of the Stanley order compared: they agree on all 203,455 ordered pairs for $2 \le n \le 8$. As an external check, the right-depth vectors for $n = 5$ were shifted by $c_j = r_{j+1} - 1$ and compared with the fourteen depth sequences printed in Fig. 41 of [14]: the two sets are equal. The bounds of Proposition 6.3, the threshold of Lemma 8.1 and the basic inequalities of Lemma 5.2 were verified on the ranges where exact integers remain feasible, with no exceptions.

Independently, the order was computed for $n \le 6$ by evaluating all trees at several thousand assignments and comparing logarithms in high precision. That last computation is experimental only: floating-point logarithms carry no certificate for the sign of a comparison, and it should be read as evidence rather than proof. Its outcome is that the number of comparable pairs agrees with the number of Stanley-comparable pairs (9 at $n = 4$, 70 at $n = 5$), and all 309 Stanley-incomparable pairs at $n = 6$ were separated in both directions. The shape of that agreement is worth stating, but so is its limit. For a finite set $A$ of assignments write $T \le_A U$ when the comparison holds at every member of $A$; then $\le_{\mathrm{St}} \subseteq \le_{\mathrm{ev}} \subseteq \le_A$, the second inclusion because a finite probe can only *enlarge* the relation. So if $\le_A$ were computed with certified comparisons and came out equal to $\le_{\mathrm{St}}$, the three relations would coincide at that $n$. **That is not what the floating-point run provides**: an uncertified sign gives no such guarantee, and the agreement reported above is evidence and nothing more. The proofs of Theorem 1.1 and Corollary 1.3 do not use it; the certified counterpart is Corollary 1.3 itself, whose $n - 2$ assignments are exactly such a set $A$.

**Example 15.1.** For the pair of Example 4.3, Theorem 1.2 gives the two assignments explicitly. With leaves $(2, 2, 2, 512)$ one has $\mathrm{ev}_3\big((ab)(cd)\big) = 2^{2^{513}}$ and $\mathrm{ev}_3\big((a(bc))d\big) = 2^{2048}$;

with leaves $(2, 2, 128, 2)$ one has $\mathrm{ev}_3\big((ab)(cd)\big) = 2^{32768}$ and $\mathrm{ev}_3\big((a(bc))d\big) = 2^{2^{129}}$. Both inequalities are visible without computing either number.

The statements of Part II were checked in the same spirit: the five bracketings of Example 3.1 and the collisions it produces were computed exactly, the equality classification of Theorem 1.10 was tested against $\mathrm{H}_r$ for $3 \le r \le 5$ over the ranges where the values remain representable, and the boundary example $(a, b, c, n) = (2, 2, 1, 4)$ for the standalone wording of Barrett's Theorem 6.6 was computed both numerically and by hand before being confirmed in the original; how it relates to the exclusion of 0 and 1 made across his section is discussed in Section 17.

# 16 The Lean development

## 16.1 Part II: the hierarchy

The development is written for Lean 4 (v4.30.0) [16] with Mathlib (v4.30.0) [18], at the revision `c5ea00351c28` pinned in the manifests of both developments. Its public project page is

`https://github.com/turahigashi/hyper-tamari`

It consists of six files in the directory `HyperTamari/` — `Basic.lean`, `Tetration.lean`, `General.lean`, `Nested.lean`, `VarLeaves.lean` and `Stanley.lean` — together with the raw `lake build` output recording the axiom audit (`logs/axiom-audit.txt`). The version described here is the one tagged `v1.15.0-pre` in that repository, and a snapshot of that tag is archived at `https://doi.org/10.5281/zenodo.22789946`.

The correspondence with the numbered statements of this paper is given in Table 1. The population of the table is fixed as follows. It consists of every numbered environment of Section 3 and of Sections 10–13; of those stated in Section 1 apart from Theorems 1.1 and 1.2 and Corollary 1.3, which are the headline statements of Part I; and of the six environments of Section 2 that Part II uses (Definitions 2.1, 2.3 and 2.4, Lemmas 2.2 and 2.8, and Proposition 2.9). There are 24 of them: 6 theorems, 3 propositions, 6 lemmas, 2 corollaries, 3 definitions, 1 example and 3 remarks. The environments of Section 2 that belong to Part I (Definitions 2.5 and 2.6) and the statements of Part I are covered instead by the second development described below. Twenty-three of the 24 appear in the table. The one that does not is Remark 1.12, which relates the two parts and has no counterpart in the Part II development. The only other row is one recording the specialisation of the labelled-tree results to a single repeated label. Section 14 and Lemma C.1 of Appendix C also belong to Part II and are also formalised, but they were written after this table and are described in the text at the end of this section rather than given rows here. The population and the two exceptions are checked by `check_table.py` against the source of this paper; the existence of every declaration name printed in the table, and in this section, is checked separately by `check_names.py` against the audit logs and the Lean sources. Neither script verifies that a named declaration has the mathematical type claimed for it beside its name. A numbered statement with several clauses may occupy several rows. We distinguish three cases, because they carry different evidential weight:

EXACT a single declaration with the same quantifiers, hypotheses and conclusion as the printed clause;

COLLECTIVE the clause is covered by a combination of declarations (typically because the paper states several things at once);

PROSE background material that is *not* formalised here — standard facts about the Tamari lattice that we use as context, or explanatory remarks.

We stress the third row of this classification: the paper does *not* claim that everything it says is machine-checked. For Part II the table records, clause by clause, the declarations supporting the numbered statements that assert an inequality or an equality about $\mathrm{H}_r$ or about the evaluation of trees; but its coverage must be read together with the three limits listed below, and in particular the passage from equality at the root to equality in an arbitrary context is not supplied by the declarations listed. Background statements marked PROSE are outside the development altogether. For Part I the coverage is smaller, and the exceptions are named below.

The table itself is Table 1, printed in Appendix A so as not to interrupt the account.

Two assertions of Section 12 are not numbered and therefore lie outside the population of the table; both are formalised: the sharpness statement for all ranks ("for every $r \geq 3$ and all $x, z \geq 2$, taking $y = 1$ reverses the inequality") is `ht_reversed_at_y_one`, and the equality of both sides at $x = 1$ is `ht_eq_at_x_one`. The value pair $256 < 65536$ quoted in Section 11 is `ht_rank4_strict_smallest`. The development also contains `pow_lt_pow_right'` and `pow_right_injective'`, choice-free replacements for the corresponding Mathlib lemmas (see below), and the auxiliary monotonicity facts `succ_le_hyper`, `hyper_mono_arg`, `hyper_mono_base`, `hyper_rank_mono`, `hyper_lt_arg` and `hyper_lt_base`.

No declaration uses `sorry`, `native_decide` or any private axiom. Every one of the **156** theorems in these six files is audited with `#print axioms`, and `scripts/check_audit.py` checks mechanically that the audit covers them all: declared 156, audited 156, missing 0. `sorryAx` does not occur. The axiom dependencies are `propext` alone (5 theorems), `propext, Quot.sound` (139), and twelve theorems that depend on no axioms at all. In particular

**no declaration depends on `Classical.choice`.**

This is not a foundational claim, and choice-freeness is not what the paper is about: every statement here is an elementary assertion about natural numbers, and nobody would expect it to need choice. We record it because it costs nothing and makes the artifact reusable — a development free of `Classical.choice` can be transported into constructive settings, or used inside a larger development that must itself avoid choice, without reproving anything.

It is worth saying where the choice had been hiding, since it was not in the mathematics. An earlier version of this development did depend on `Classical.choice` in 30 of its declarations, and the dependency was traced to exactly four sources, all of them incidental: the Mathlib lemmas `Nat.pow_lt_pow_right` and `Nat.pow_right_injective` (whose companion `Nat.pow_le_pow_right` is proved with no axioms at all), the `nlinarith` tactic, and — the one that is easiest to miss — `norm_num` applied to a numeric *inequality*, which pulls in choice, while `omega` discharges the same goal without it and `norm_num` applied to an *equation* does not. Replacing these by `omega`, by `decide`, by `Nat.add_le_mul`, and by two short inductions built on the axiom-free `Nat.pow_lt_pow_succ` removed the dependency entirely, with no change to any statement.

### 16.2 Part I: rank three

The arguments of Sections 5, 6 and 8 are formalised in a second Lean 4 development, also on top of Mathlib, which is in the directory `part1/` of the same repository. Section 7 is formalised in the Part II development instead, because the covering machinery it uses lives there. The formalisation covers Lemmas 5.2, 5.3, 5.4 and 8.1, Definition 6.1, Proposition 6.3, Theorem 1.2, and the implication $T \leq_{\mathrm{ev}} U \Rightarrow T \leq_{\mathrm{St}} U$ of Theorem 1.1. Evaluation is defined at an arbitrary list of leaf values, and a separate lemma identifies the value at the

assignment "leaf $i$ carries $t$, all other leaves carry 2" with the quantity that Proposition 6.3 bounds, so the formalised statement is about the evaluation order itself. An audit enumerates every declaration from the Lean environment and reports what each depends on: of 292 declarations, some of them generated by Lean from the inductive types, 105 depend on no axiom at all and the remaining 187 only on propext and Quot.sound; none uses the axiom of choice, and none contains a hole; the raw output is kept with the artifact as `part1/logs/axiom-audit.txt`. The examples above are part of the formalisation, including the two thresholds 512 and 128 and the comparison with Fig. 41 of [14].

The uniform bound of Section 9 is formalised as well: Lemmas 9.1, 9.2 and 9.3, the auxiliary Lemma 2.7, and the substantive direction of Corollary 1.3, namely that the $n-2$ displayed inequalities force $T \leq_{\mathrm{St}} U$ (`Tree.stanleyLe_of_testFamily`). Theorem 7.1, the other direction, is formalised in the Part II development, where the cover machinery lives (`StRot.eval_le_rank3`).

**What the two halves of Theorem 1.1 are, exactly.** The forward implication is machine-checked end to end and in one place: `Tree.stanleyLe_of_evalLe` takes `Tree.evalLe`, which quantifies over *every* assignment of leaf values $\geq 2$, and returns `Tree.stanleyLe`. The converse is not a single declaration. Part I proves that the Stanley order is exactly the reflexive-transitive closure of the covers (`Tree.stanleyLe_iff_reflTransGen_stRot`, with `Tree.stCover_iff_stRot` identifying the covers), and Part II proves that evaluation does not decrease along such a chain (`StRot.eval_le_rank3_star`). **What is not formalised is the join**: the two developments use different tree types — Part I's unlabelled `Tree` with an assignment given as a list, Part II's `LTree` with the values at the leaves — and the translation between them, and the passage from one evaluation to the universally quantified `evalLe`, are done on paper here and not in Lean. So the converse is machine-checked in two pieces over two representations, and the step that joins them is carried out here on paper. **We do not claim a kernel-checked proof of the equivalence of Theorem 1.1 as a single statement**, and a reader who wants one should regard the converse direction as proved by the text, with Lean checking its two ingredients.

**What is not machine-checked.** Three things for the main results, and the artifact says so; the background material marked PROSE in the table is outside the scope as well.

1. The joins between the two developments: the one just described for Theorem 1.1, and, for Theorems 7.1 and 14.3, the identification of the local move on Part I's unlabelled trees, which Part I proves to be the Stanley cover, with the move on Part II's labelled trees along which Part II proves evaluation monotone. Neither join is a theorem of either development.

2. Corollary 7.2, the classification of equality along a Stanley cover at rank three. Part II has the scalar identity (`ht_rank3_eq_iff`) and the equality condition for a Tamari rotation at the root (`LRot.root_eq_rank3_iff`); neither is the statement for a general Stanley cover in a context.

3. The rank-three contextual consequences of Corollary 13.2, including strictness for a repeated label $a \geq 3$ at an arbitrary node. The declarations listed prove equality and strictness *at the root*; the propagation through a context is supplied on paper (Appendix B). High-rank strictness in an arbitrary context is formalised.

Proposition 4.1 is formalised as `Tree.stDom_iff_stanleyLe`, in the convention in which Definition 2.5 states it: for two trees with the same number of leaves, $H_{w(T)}(k) \leq H_{w(U)}(k)$ for every $k$ if and only if $r_i(T) \leq r_i(U)$ for every $i$. Each of the four ingredients of its printed proof is a separate declaration: the words $q$ and $w$ (Definition 2.5) with the computation that the height after the $j$th up-step of $qT$ is the $(j+1)$st

right depth (`Tree.upHeights_eq_rseq_tail`); the arithmetic relation $p_j + e_j = 2(j + 1)$ between position and height (`Tree.upPos_add_upHeights`); the counting duality, in `Word.countP_le_iff_getElem_le`, between prefix counts and up-step positions; and the passage from $q$ to $w$, which is the identity $w = \overline{q}$ (`Tree.wword_eq_barW_qword`) together with $H_{\bar{u}}(k) = H_u(N - k)$ for a balanced word (`Word.ht_barW`).

Lemma 4.4 is formalised as well, together with Lemma 4.5, in the file `ChowStanley/Cover.lean`. There `Tree.StCover` is the order-theoretic cover — $T$ below $U$, not conversely, and no tree strictly between, quantified over *all* trees — and `Tree.SwapDU` is the word statement of Lemma 4.4, that $w(U)$ is $w(T)$ with one occurrence of $DU$ replaced by $UD$. The two are proved equivalent (`Tree.stCover_iff_swapDU`), and so is the local form of Lemma 4.5 (`Tree.stCover_iff_stRot`, where `Tree.StRot` is the inductively defined move $((A \cdot B) \cdot R) \mapsto (A \cdot L_{\mathbf{D}}(B \cdot c))$ that the Part II development also uses). Along the way two further facts are proved that the paper otherwise cites: that the right-depth sequence determines the tree (`Tree.rseq_injective`, which is 2.8 of [4]) and that the word determines it too (`Tree.wword_injective`).

The ingredient the printed proof of Lemma 4.4 does not need, and which the formalisation has to supply, is that raising one coordinate of a right-depth sequence by one again gives the sequence of a *tree*. In the literature this is stated at the end of [4, 2.8] — "*every zag sequence is the RD-sequence of some bracketing*", a zag sequence being one with $d_1 = 0$, $d_2 = 1$ and $1 \le d_{i+1} \le d_i + 1$ — and we could have formalised that. Instead, because only one raise at a time is needed, the development performs surgery on $T$ itself: `Tree.raise` descends to the place where the raise happens and applies the move of Lemma 4.5 there (`Tree.rseq_raise`). Non-degeneracy is checked — a cover exists, its word comparison really is $UDUD \mapsto UUDD$, and the bottom and top of the four-leaf order are comparable without being a cover.

In the Part II development, the clause of Corollary 13.2 that every cover is strict unless $(r, a) = (3, 2)$ is formalised for rotations at the root; the general-position case is the printed argument.

Lemma C.1 is formalised in the file `HyperTamari/Nested.lean` (11 of the 156 theorems). Its statement is `nested_bound_strict`, for every $r \ge 4$, $x \ge 2$ and $y, z \ge 1$, and the proof is the one of Appendix C: the rank-four invariant (4) is `nested4_aux`, the step from rank $r$ to rank $r + 1$ is `nested_step`, and the non-strict form (2) is `nested_bound`.

Section 14 is formalised in the file `HyperTamari/Stanley.lean` (29 of the 156 theorems). There the Stanley covering relation is taken in the local form of Lemma 4.5; that this is exactly the covering relation of the Stanley order, which is Lemma 4.4, is proved in the Part I development for the trees the order lives on, and the relation used here is the same one with labels carried along, an identification made on paper (the first item of the list above). The theorem `StRot.eval_lt` **has no hypothesis beyond** $r \ge 4$ **and labels** $\ge 2$: its arithmetic input is `nested_bound_strict`, so it follows the shorter route of Remark 14.4 rather than the printed proof, which splits at $m = 0$. Everything in that file is proved: that grafting on the left computes the substituted value, that $G$ is strictly increasing, the estimate $b + G(c) \le G(b * c)$, and the propagation through a tree context. Witnesses are included: the pair $((2 \cdot 2) \cdot (2 \cdot 2))$ and $(2 \cdot ((2 \cdot 2) \cdot 2))$, which is a Stanley cover but *not* a Tamari one, is exhibited with its rank-3 values 256 and 65536, and four instances of the nested bound are computed at rank 4 directly from the definition.

Every Lean name printed in this section *and in the table of Appendix A* — 109 of them — is checked against the two developments by `check_names.py`. It reads fully qualified names from the two audit logs and from the sources, so that a namespace printed here has to be the declaration's actual namespace; the five Mathlib names quoted above are checked by elaborating them against Mathlib. **What that script does not do** is compare what a declaration *says* with the printed statement beside it: it checks existence, not agreement.

The formalisation is a check on the arguments printed here, not a substitute for them.

# 17 Related work, and what we claim

Csákány and Waldhauser [4] introduce the associative spectrum and show that exponentiation is Catalan: for pairwise distinct prime leaves, distinct bracketings give distinct values. Their concern is the number of values; the order in which those values arrange themselves is not addressed: Tamari's paper [23] and a paper of Bennett and Birkhoff on Newman lattices appear in their bibliography only as general references for bracketings, and no order on bracketings is used. Huang and Lehtonen [12] extend the invariant to the *associative-commutative spectrum* and determine it for exponentiation (their Proposition 6.1.5): the associative spectrum of exponentiation is $C_{n-1}$ and its ac-spectrum is $n^{n-1}$. Their question is again how many distinct term operations the bracketings produce; ours is how the resulting values are *arranged*, and neither the Tamari order nor any statement comparing values along an order occurs in that line of work. Huang, Mickey and Xu [13] compute the associative spectrum of a further operation. Earlier, Göbel and Nederpelt [6] counted the values of the bracketings when every leaf carries the same real number $x > 1$: there are at most as many as there are rooted trees with one edge fewer than the number of leaves, with a sharper upper bound for $x = 2$ in terms of trimmed trees, and, for each number of leaves, exactly that many for all but countably many $x$ (their Theorems 2.2, 3.1 and 5.1); they, too, count values and do not compare them. Tamari [23] enumerated the bracketings, and Friedman and Tamari [5] proved that the right-rotation order on them is a finite lattice (the statement, they note, is already in Tamari's thesis of 1951, without proof) — their title already calls the relation a *loi demi-associative*, which is precisely the law that runs through the whole hyperoperation hierarchy. Zeilberger [25] gives a sequent calculus for that law. Gryszka [9] studies the comparison of power towers by normalising them, which concerns right-associated towers rather than general bracketings.

## 17.1 Where the rank-three case comes from, and what we add

It is important to say clearly what part of Proposition 1.6 is already known. In a MathOverflow question of January 2018 [19], Spivak treats exponentiation on the integers $\geq 2$ as an ordered magma and orders bracketings by the *evaluation order*: $E \leq F$ when the inequality holds *for every* assignment of values to the leaves. He asks how this order relates to the height (Stanley) order on Dyck paths. In the answers, Chow, following a comment of Rubey on the question, proves that a Stanley cover does *not* decrease the value, the computation resting on

$$B^{CD} = (B^C)^D \;\geq\; (BC)^D \;\geq\; BC^D \qquad (B, C \geq 2,\; D \geq 1),$$

and Rubey, independently, reduces the same covering relation to the inequality $L_2^{(L_1^{xR})} \geq (L_2^{L_1})^{(x^R)}$, which he leaves unverified. That inequality is not $(\mathrm{HT}_r)$ at rank 3: writing it as $x'^{(y'^{xR})} \geq (x'^{y'})^{(z')}$ with $z' = x^R$, it reduces to $(\mathrm{HT}_r)$ only when $xR = x^R$, which for $x \geq 2$ and $R \geq 1$ happens exactly when $R = 1$ or $(x, R) = (2, 2)$; outside those it is strictly stronger, being the inequality attached to a *Stanley* cover rather than a Tamari one.

Since the evaluation order quantifies over *all* leaf assignments it is stronger than the single-label evaluation $\mathrm{ev}_{3,a}$, and since the Stanley lattice is an extension of the Tamari lattice, comparability in the Tamari order implies comparability in the Stanley order. Consequently **the rank-3 monotonicity is already contained, in a stronger form, in that discussion, and we do not claim it.**

That discussion is, however, explicitly unfinished, and it is confined to rank 3. Chow states that he can "complete half of the proof", the half we have just quoted; the converse

half — that incomparable bracketings can be separated in both directions by a suitable choice of leaf values — he leaves open, writing "I haven't quite figured out how to say this rigorously". Rubey likewise reduces the covering relation to the displayed inequality and stops there. In comments under his answer he suggested that Chow's idea should also give the converse, first by a reduction to paths that differ in one valley and one peak and then, after Chow cautioned that reductions for this half need care, by comparing two auxiliary paths that separate at a single peak, with "some details to check here"; the sketch was not completed. Neither treats any rank above 3, and neither asks when equality holds.

Theorem 13.1 above takes up the first half at every rank. It says that for arbitrary leaf values $\geq 2$ the evaluation is monotone along the Tamari order at every rank $r \geq 3$, and *strictly* monotone from rank 4 on. **Chow's converse half is settled at rank 3 by Theorem 1.1 of Part I**; above rank 3 it remains open, and Section 18 says what would be needed.

There is also a body of peer-reviewed work on orders attached to iterated exponentials, and it is worth saying precisely how it relates to the present question, because the titles suggest a closer overlap than there is. Brunson [3] fixes $e \leq x_1 \leq \cdots \leq x_n$ and, *for each permutation* of the $x_i$, forms the tower $T(y, z, \ldots, w) = y^{z^{\cdots^{w}}}$, in which "association is always to the upper right"; the resulting partial order lives on the symmetric group $S_n$. Brunson's order agrees with the Bruhat order for $n \leq 4$, up to a duality that depends on the conventions, and Brunson conjectured properties that the Bruhat order has; that the two orders nevertheless differ was shown independently, and at about the same time, by Stembridge [22], who gave an explicit characterisation of the order that disproves those conjectures, and by Griggs–Wachs [11], whose counterexample at $n = 5$ also disproves Brunson's Conjectures A and B. (The independence and the near-simultaneity are stated by Griggs and Wachs themselves.) Griggs [8] studies the poset of towers over the two-letter alphabet $\{a, b\}^n$, the tower again being "evaluated in the conventional way, from top down".

In every one of these the *bracketing is fixed* — it is the right comb — and what varies is the assignment of entries to its leaves. In the single-label problem of Sections 2–12 the situation is exactly the reverse: the entries are fixed, all equal to $a$, and what varies is the bracketing. (Section 13 then lets the entries vary as well, but still over *all* bracketings rather than over the right comb alone.) The two families of questions are therefore orthogonal, and none of these papers contains the inequality studied here. (None of [3, 8, 11] contains an occurrence of the words *bracketing*, *Tamari*, *binary tree*, *Catalan*, *rotation* or *Dyck*.)

## 17.2 Nested upper bounds in the hyperoperation literature

The inequality $(\mathrm{HT}_r)$ compares a nesting on the left with a nesting on the right. A *stronger* upper bound, in which the right-hand side carries the sum $y + z$ — which is at most $\mathrm{H}_r(y, z)$ — in place of $\mathrm{H}_r(y, z)$, has been recorded before, and at every rank. Leonardis, d'Atri and Caldarola [15] state, as their Lemma 4.8,

$$(A \uparrow^k B) \uparrow^k C \;\leq\; A \uparrow^k (B + C) \qquad (k \geq 2),$$

with the tetration case $k = 2$ isolated as their Lemma 4.6; in the notation of Section 2 this is $\mathrm{H}_r(\mathrm{H}_r(x, y), z) \leq \mathrm{H}_r(x, y + z)$ for $r \geq 4$. In the published version the proof of Lemma 4.8 assumes $A \geq 3$ and leaves the case $A = 2$ to the reader; Appendix C says why that case needs an argument, and gives one. In an earlier preprint, Saibian [20] states the same comparison with a *strict* inequality (his Theorem I), and proves it: he sets down nine axioms, derives from them that each $\mathrm{H}_r(b, \cdot)$ is strictly increasing and exceeds its argument and that each rank dominates the one below, proves the tetration case $(b \uparrow\uparrow m) \uparrow\uparrow n < b \uparrow\uparrow (m + n)$ for $b > 1$, and lifts it to every higher rank by induction.

This matters for the present paper, because $(\mathrm{HT}_r)$ follows from either of them in two lines. For positive integers $y, z \geq 2$ one has $y + z \leq yz \leq \mathrm{H}_r(y, z)$, and $\mathrm{H}_r(x, \cdot)$ is nondecreasing, so

$$\mathrm{H}_r\big(\mathrm{H}_r(x, y), z\big) \;\leq\; \mathrm{H}_r(x, y + z) \;\leq\; \mathrm{H}_r\big(x, \mathrm{H}_r(y, z)\big). \tag{3}$$

With Saibian's strict form the first inequality is strict, so (3) already yields the strictness of Theorem 1.10(3) on the range $z \geq 2$; the remaining boundaries $z = 0$ and $z = 1$ are immediate from the definition. **We therefore do not claim Theorem 1.9, nor the high-rank half of Theorem 1.10, as new arithmetic**; our proofs in Section 12 are independent, but the results are not.

One caveat about two of these sources should be stated plainly: we cite Saibian [20] in its viXra version and Barrett [2] in its Preprints.org version, rather than as refereed journal articles. We have read both in the original and checked the statements we quote — Saibian's Theorem I and Barrett's Theorems 6.5 and 6.6. Neither source is a bare announcement: each carries a proof, and we have followed the structure of Saibian's, although we have not audited the whole of his thirty-two pages. **No result of this paper rests on either of them.** They are cited for attribution: the arithmetic they contain is proved here as well, in Section 12 and Appendix C, and a reader who declines to rely on preprints should read those as the proofs of Theorem 1.9, Theorem 1.10 and Lemma C.1 rather than as second proofs of them. **Nor does any result rest on an outside proof of a nested bound.** The one such bound that Theorem 14.3 uses is the non-strict inequality (2) of Leonardis, d'Atri and Caldarola [15], which appeared in a refereed journal; but it is used at the base 2, which their published proof does not treat, and so it is proved here as Lemma C.1. In the Lean development the older, shorter derivation survives, through the strict form of that lemma, which is proved there as well.

The equality question has also been addressed directly. Barrett [2], in a preprint on factorisation under hyperoperations, proves as his Theorem 6.5 that $(a^b)^c = a^{(b^c)}$ with $a, b, c > 1$ forces $b = c = 2$; this is exactly the non-trivial rank-3 equality of Theorem 1.4. His Theorem 6.6 asserts that for $n \geq 4$ and $a > 1$, equality in $(a[n]b)[n]c = a[n](b[n]c)$ forces $b = c = 1$. Read as a statement on its own this is defective at the boundary: taking $c = 1$ gives

$$\mathrm{H}_n\big(\mathrm{H}_n(a, b), 1\big) = \mathrm{H}_n(a, b) = \mathrm{H}_n\big(a, \mathrm{H}_n(b, 1)\big)$$

for all $a, b \geq 2$ and $n \geq 2$, so that $(a, b, c, n) = (2, 2, 1, 4)$ is an equality with $b = 2$; both sides equal 4. (The corollary invoked in that proof requires all four compared arguments to exceed 1, which fails at $c = 1$.)

**This is not a refutation of Barrett's result.** The opening of his §6.2 says explicitly that the discussion will "also ignore cases where any of the operands is 0 or 1", and the sentence closing the proof of Theorem 6.6 returns to the conclusion that there is no non-trivial associativity for $n \geq 4$ with $a, b, c > 1$. On that domain his Theorem 6.6 *is* Theorem 1.10(3). What we add is not a correction of his mathematics but a single statement that carries its own boundary conditions: Theorem 1.10 says, for $x, y \geq 2$ and $r \geq 4$, that equality holds if and only if $z = 1$, and it settles $z = 0$ as well. The hypotheses $x, y \geq 2$ are needed: at $r = 4$, $x = 2$, $y = 0$, $z = 3$ both sides are 1.

A remark on the minimiser is also in order. The left comb at rank 3 evaluates to $\mathrm{ev}_{3,a}(L_n) = a^{a^{n-1}}$, and this is the operation known in the hyperoperation literature as the *lower* hyperoperation $\mathrm{lower}_4$, obtained by iterating exponentiation to the left rather than to the right; see Andonov [1], who places it in the lower hyperoperation sequence. The identification is with the rank-3 left comb specifically: the left-iterated operations of higher rank are not in general the left combs of the corresponding rank, so we do not extend the name upwards.

### 17.3 The associative spectrum, and the two uses of right depth

The bracketings of a word under a fixed binary operation have been studied as the *associative spectrum*, the sequence counting how many distinct term operations they induce. Csákány and Waldhauser [4] introduced the invariant and computed it for exponentiation in 3.4, obtaining the Catalan numbers; Lehtonen and Waldhauser [17] determine the spectra of graph algebras; Huang and Lehtonen [12] introduce the associative-commutative spectrum and compute it for exponentiation in their Proposition 6.1.5. That line of work counts how many operations the bracketings produce; the present question is how the resulting values are *arranged*. The two are different, and the difference is exactly the gap this paper closes: knowing that two bracketings induce different functions says nothing about which one is the larger at a given assignment, let alone at every assignment.

A separate body of work orders *towers* rather than bracketings. Brunson [3] fixes the right-nested bracketing and permutes the entries; Stembridge [22] and Griggs–Wachs [11] compare the resulting order with the Bruhat order. There the bracketing is fixed and the entries move; here the entries are arbitrary and the bracketing moves.

### 17.4 What we do claim

Not finding a result is not a proof that it is new, so we state the claim in pieces, and we begin with what is *not* claimed.

**Not claimed, Part I.** The inclusion $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ at rank three is due to Chow [19]; Rubey independently reduced it to a single inequality. We reprove it as Theorem 7.1, by a different and shorter argument, only so that the paper is self-contained; the result is his. The coordinate presentation of Proposition 4.1 is known: Knuth defines the Stanley lattice by depth coordinates [14], Lehtonen and Waldhauser identify DFS depth with right depth [17], and 2.8 of [4] proves that right-depth sequences determine a bracketing. The reading of $r_i$ as the level of a leaf in the resulting tower, which is the qualitative content of Proposition 6.3, is 3.4 of [4]. The idea of enlarging one variable to force a comparison is Chow's, stated in the same answer [19].

**Not claimed, Part II.** The rank-3 monotonicity is known [19]. The local law ($\mathrm{HT}_r$) at every rank, and its strictness for $r \geq 4$, follow in two lines from the nested upper bounds of Leonardis–d'Atri–Caldarola [15] and Saibian [20] by (3); Theorem 1.9 and parts (1), (3), (4) of Theorem 1.10 are therefore *not* new arithmetic, and our independent proofs in Section 12 are offered as proofs, not as first occurrences. The nested bound of Lemma C.1 is theirs as well; we prove it only because the published proof of [15, Lemma 4.8] does not treat the base 2. The non-trivial rank-3 equality $(y, z) = (2, 2)$ of Theorem 1.4 is Barrett's Theorem 6.5 [2]; its instance $x = y = z = 2$, the relation $(2^2)^2 = 2^{(2^2)}$, is the one in view of which Göbel and Nederpelt [6] expect the base 2 to behave differently.

**What we offer.** First, and principally, the converse inclusion, and hence Theorem 1.1: at rank 3 the evaluation order is exactly the Stanley order, which settles the half of [19] that was left open. Second, the quantitative form of that separation, Theorem 1.2, in which the leaf to enlarge, the value to put there, and the fact that all other leaves may remain at 2 are determined explicitly, together with the finite test family of Corollary 1.3. Third, the equality question settled by a single statement carrying its own boundary conditions: Theorem 1.10 classifies equality at *every* rank and at every boundary, where the statement of Barrett's Theorem 6.6 taken on its own is defective at $z = 1$ although his section excludes that case by hypothesis. Fourth, Theorem 14.3: the previously stated *non-strict* nested bound, together with the two-unit gap of Lemma 14.1, is strong enough to carry the Stanley covering relation, so evaluation is strictly Stanley-monotone at every rank $r \geq 4$ — which

settles one of the two implications of Conjecture 18.1 at every rank above 3. Fifth, the organisation of the rest of that arithmetic as an order-theoretic statement about the Tamari lattice: Proposition 1.6, Theorem 1.7 and Corollary 1.8 locate the failure of strictness at the single pair $(r, a) = (3, 2)$ and show that from rank 4 on the evaluation is strictly increasing along every cover, and Theorem 13.1 gives the cover version for arbitrary leaf values $\geq 2$. Sixth, the Lean formalisation, whose extent and whose three limits are set out in Section 16: it covers the forward implication of Theorem 1.1 end to end, both of the ingredients of the converse, and Lemmas 4.4 and 4.5 with 2.8 of [4].

Within the literature we have searched we have not found the converse inclusion, the quantitative separation, the Tamari formulation, the complete equality classification, or an equivalent formalisation; we do not convert that into a claim of first occurrence. We should add that the two ingredients of Part I that we take as known were found late, in the course of checking the literature for this paper, and that they are close to the result: 2.8 and 3.4 of [4] together give the injectivity of $r$ and the interpretation of $r_i$ as a level in the tower, which is most of the apparatus used in Sections 4–8. What they do not give is any comparison of magnitudes: substituting distinct primes and appealing to unique factorisation shows that two values differ, never which is larger. We should also say that the qualitative converse — that *some* sufficiently large value at leaf $i$ separates — can be reached from their apparatus by a short asymptotic argument: fixing every other leaf at 2 and letting $t \to \infty$, an induction along the path to the root gives $\log^{\circ h} v_{T,i}(t) \sim Ct^{\alpha}$ with $h = r_i(T)$ and $C, \alpha > 0$, so a tree with the larger right depth wins for all large $t$. That argument gives no threshold. What Theorem 1.2 and Corollary 1.3 add is the threshold and the uniformity: which leaf, what value, and a finite family of assignments fixed in advance.

## 18 What remains open

The two parts fill two corners of a square whose axes are the rank and the order.

| | **Tamari** | **Stanley** |
|---|---|---|
| rank 3 | $\leq_{\mathcal{T}} \subseteq \leq_{\mathrm{ev}}$, equality classified | $\leq_{\mathrm{St}} = \leq_{\mathrm{ev}}$ (Theorem 1.1) |
| rank $\geq 4$ | $\leq_{\mathcal{T}} \subseteq \leq_{\mathrm{ev}}$, strict on covers | $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ (Theorem 14.3); converse **open** |

The entries in the first column are containments only. We have not proved that either is proper directly: strict increase of the *values* along a cover does not make the containment of *relations* strict. Properness does follow, for $n \geq 4$ and at *every* rank, from the second column together with $\leq_{\mathcal{T}} \subsetneq \leq_{\mathrm{St}}$; for $n \leq 3$ the two orders coincide, since there are at most two trees and they are Tamari-comparable.

What is missing is one implication, the same one at every rank above 3.

**Conjecture 18.1.** *For every $r \geq 4$ the evaluation order $\leq_{\mathrm{ev}}$ on $\mathcal{T}_n$, taken over all leaf assignments with values $\geq 2$, is the Stanley order.*

At rank 3 the same statement is Theorem 1.1. One half of Conjecture 18.1 is Theorem 14.3: the inclusion $\leq_{\mathrm{St}} \subseteq \leq_{\mathrm{ev}}$ holds at every rank $r \geq 4$, and in the strict form. What is missing is the converse $\leq_{\mathrm{ev}} \subseteq \leq_{\mathrm{St}}$, and the obstruction is easy to name: at rank 3 the converse is Theorem 1.1, which rests on Proposition 6.3, whose two-sided estimate is built from the collapsing law $(x^y)^z = x^{yz}$ through the absorption Lemma 5.3. No analogue of that identity is known for tetration, and without it we cannot bound from above the tower that one enlarged leaf generates.

So at rank $\geq 4$ the evaluation order is known to contain $\leq_{\mathrm{St}}$ and is therefore not strictly between $\leq_{\mathcal{T}}$ and $\leq_{\mathrm{St}}$; whether it is exactly $\leq_{\mathrm{St}}$ or strictly larger, we do not know.

# A The correspondence table for Part II

This is the table described in Section 16; its population, and the two statements that do not appear in it, are fixed there. The row for Corollary 13.2 supplies the root-level ingredients only: the rank-three contextual consequences are proved on paper, not by the declarations named. The last sentence of Corollary 1.8 carries the same limitation. Entries marked PROSE are not formalised.

Table 1: Correspondence between the 23 numbered statements of Part II (Sections 3 and 10–13, with Sections 1–2) and the Lean development. Clauses of one statement are listed on separate rows.

| Statement | Lean declaration(s) | Status |
|---|---|---|
| Thm. 1.4 (inequality) | `ht_rank3` | EXACT |
| Thm. 1.4 (equality iff) | `ht_rank3_eq_iff` | EXACT |
| Thm. 1.4 (sharpness of $y \geq 2$) | `ht_reversed_at_y_one` at $r = 3$ (universal), `ht_needs_two_le_y` (the witness $x = z = 3$) | COLLECTIVE |
| Thm. 1.5 (inequality) | `tet_ht`, `ht_rank4` | EXACT |
| Thm. 1.5 (strict for $z \geq 2$) | `tet_ht_strict_of_two_le`, `ht_rank4_strict_two` | EXACT |
| Prop. 1.6 (covers) | `Rot.eval_le` | EXACT |
| Prop. 1.6 (Tamari order) | `Rot.eval_le_of_reflTransGen` | EXACT |
| Prop. 1.6 (right comb) | `rightComb`, `eval_rightComb` (with $n \mapsto n - 1$) | COLLECTIVE |
| Thm. 1.7 (covers) | `Rot.eval_lt` | EXACT |
| Thm. 1.7 (Tamari order) | `Rot.eval_lt_of_transGen` | EXACT |
| Cor. 1.8 | `Rot.eval_le`, `rank3_rotation_not_strict`, `Rot.eval_lt` | COLLECTIVE |
| Thm. 1.9 | `ht_general` | EXACT |
| Thm. 1.10 (1)–(4) | (1) `ht_eq_at_one`; (2) `ht_rank3_eq_iff`, with `ht_rank3_equality_witness` as the witness; (3) `ht_strict_general`; (4) `ht_lt_at_zero`. The summary declaration `equality_classification` packages weaker forms of these — a rank-three instance in place of (2), and $\neq$ in place of the strict inequalities of (3) and (4) | COLLECTIVE |
| Rem. 1.11 ($\mathrm{H}_r(x, y) > x$; $\mathrm{H}_r(2, 2) = 4$) | `self_lt_hyper`, `hyper_two_two` | COLLECTIVE |
| Rem. 1.11 (the two coordinates compete) | — (explanatory) | PROSE |
| Def. 2.3 | `LTree`, `LTree.eval`, `LTree.AllLeavesTwoLe`, `LRot` | EXACT |
| Thm. 13.1 | `LRot.eval_le`, `LRot.eval_lt`, `LRot.eval_le_of_reflTransGen`, `LRot.eval_lt_of_transGen`, `LRot.preserves`, `LTree.two_le_eval` | COLLECTIVE |
| Specialisation to a single label | `ofBTree`, `eval_ofBTree`, `rot_ofBTree`, `Rot.eval_lt_of_var` | EXACT |
| Cor. 13.2 | `LTree.four_le_eval_node`, `LTree.eval_eq_two_iff`, `LRot.root_eq_rank3_iff`, `LRot.root_lt_rank3_of_three_le`, `rank3_root_lt_of_three_le`, `cover_strict_unless_rank3_two` (root only; see the note above) | COLLECTIVE |
| Def. 2.1 | `hyper`, `hyper_zero`, `hyper_one_zero`, `hyper_two_zero`, `hyper_succ_succ_zero`, `hyper_succ` | EXACT |

*Table 1 (continued)*

| Statement | Lean declaration(s) | Status |
|---|---|---|
| Lem. 2.2 | `hyper_one`, `hyper_two`, `hyper_three`, `hyper_one_arg` | COLLECTIVE |
| Def. 2.3 (trees, ev) | `BTree`, `BTree.eval` (not restricted to a fixed number of leaves) | EXACT |
| Def. 2.4 (rotation, Tamari order) | `Rot` (cover), `Relation.ReflTransGen Rot` (the order) | EXACT |
| Def. 2.4 (lattice, comb extremality) | — (classical [5], used as background) | PROSE |
| Lem. 2.8 | `mul_le_pow_of_two_le`, `mul_lt_pow_of_three_le` | COLLECTIVE |
| Prop. 2.9 | `mul_eq_pow_iff` | EXACT |
| Ex. 3.1 (the five values; the colliding cover) | `rank3_five_values`, `rank3_rotation_not_strict` | COLLECTIVE |
| Lem. 10.1 | `sq_tet_le` | EXACT |
| Rem. 10.2 | `sq_le_pow_fails` (the negated universal statement), `two_pow_three_lt_three_sq` | EXACT |
| Lem. 10.3 | `add_two_le_pow`, `add_three_le_pow` | COLLECTIVE |
| Prop. 12.1 (1), (2) | `sum_lemma`, `ht_of_sum` | COLLECTIVE |
| Lem. 12.2 | `add_le_hyper` | EXACT |
| Lem. 12.3 (strict inequality; equality iff $u = v = 2$; value 4) | `add_lt_hyper`, `add_eq_hyper_iff`, `hyper_two_two` | COLLECTIVE |

# B Elementary hyperoperation facts

This appendix collects the elementary inductions used in the body. None of them uses a semi-associative inequality, so nothing here is circular. Bases are integers $x \geq 2$ unless stated otherwise.

**Growth and monotonicity in the arguments.** For every $r \geq 2$ and $n \geq 1$,

$$\mathrm{H}_r(x, 1) = x, \qquad \mathrm{H}_r(x, n) \geq x, \qquad \mathrm{H}_r(x, n) \geq n + 1;$$

moreover $n \mapsto \mathrm{H}_r(x, n)$ is strictly increasing for $n \geq 0$, and $x \mapsto \mathrm{H}_r(x, n)$ is strictly increasing on integer bases $x \geq 2$ when $n \geq 1$. These are proved together by induction on $r$. At $r = 2$ they follow from $\mathrm{H}_2(x, n) = xn$; in particular $xn \geq 2n \geq n + 1$ for $n \geq 1$. Suppose they hold at rank $r$. At rank $r + 1 \geq 3$ the values at $n = 0$ and $n = 1$ are 1 and $x$. For $n \geq 1$ put $t = \mathrm{H}_{r+1}(x, n)$; then $t \geq x \geq 2$, and the hypothesis at rank $r$ gives

$$\mathrm{H}_{r+1}(x, n + 1) = \mathrm{H}_r(x, t) \;\geq\; t + 1.$$

Induction on $n$ now yields strict increase and both growth bounds at rank $r + 1$.

For strict increase in the base, fix $2 \leq x < x'$ and induct on $n \geq 1$. At $n = 1$ the assertion is $x < x'$. If $a = \mathrm{H}_{r+1}(x, n) < \mathrm{H}_{r+1}(x', n) = a'$, then $a, a' \geq 2$ and

$$\mathrm{H}_r(x, a) \;<\; \mathrm{H}_r(x, a') \;\leq\; \mathrm{H}_r(x', a'),$$

by monotonicity in the second argument and in the base at rank $r$; this is the comparison at $n + 1$. Non-strict monotonicity follows by allowing equality throughout.

**Corollary B.1** (Propagation through a context)**.** *Let $r \geq 3$ and let a tree context have all its other leaf values $\geq 2$. Then the value of the whole tree is a strictly increasing function of the value substituted at the hole, on substituted values $\geq 2$.*

*Proof.* Induct along the path from the hole to the root. Each step applies $u \mapsto \mathrm{H}_r(u, v)$ or $u \mapsto \mathrm{H}_r(v, u)$ for a fixed $v \geq 2$, and both are strictly increasing on arguments $\geq 2$ by the above. □

In particular such a context preserves and reflects equality, and preserves strict inequality; this is what is used in Corollary 7.2 and in the proof of Theorem 14.3.

**Monotonicity in the rank.** For $r \geq 2$, $x \geq 2$ and $n \geq 1$ we have $\mathrm{H}_r(x, n) \leq \mathrm{H}_{r+1}(x, n)$. For $r = 2$ this is $xn \leq x^n$, which is Lemma 2.8. For the step, let $r \geq 3$ and assume the comparison between ranks $r-1$ and $r$. At $n = 1$ both values are $x$, and if the comparison holds at $n$ then

$$\mathrm{H}_r(x, n+1) = \mathrm{H}_{r-1}\big(x, \mathrm{H}_r(x, n)\big) \;\leq\; \mathrm{H}_r\big(x, \mathrm{H}_r(x, n)\big) \;\leq\; \mathrm{H}_r\big(x, \mathrm{H}_{r+1}(x, n)\big) = \mathrm{H}_{r+1}(x, n+1),$$

the first inequality by the induction on the rank at a positive argument and the second by monotonicity in that argument. In particular $\mathrm{H}_r(x, n) \geq xn$ for $r \geq 2$ and $n \geq 1$, which is the domination used in Section 14.

**The base 1.** For $r \geq 3$ and $n \geq 0$ we have $\mathrm{H}_r(1, n) = 1$. At rank 3 this is $1^n = 1$. At higher ranks, induction on $n$ uses the value 1 at $n = 0$ and $\mathrm{H}_{r-1}(1, 1) = 1$ from Lemma 2.2. These are the boundary comparisons appealed to in Section 12.

# C The nested upper bound

The proof of Theorem 14.3 applies (2) with $x$ the value of a subtree, and that value is 2 whenever the subtree is a leaf labelled 2. The bound is Lemma 4.8 of Leonardis, d'Atri and Caldarola [15], and in strict form, for bases $b > 1$, it is Theorem I of Saibian [20]. In the published version of [15] the proof of Lemma 4.8 assumes $A \geq 3$ and leaves the case $A = 2$ to the reader; for $k = 2$ it rests on their Lemma 4.6, whose proof ends by passing from a tower built on $A + 1$ to one built on $A$, a step that needs $(A+1) \uparrow\uparrow m \leq A \uparrow\uparrow (m+1)$ with $m = B + C - 3$ and already fails at $A = 2$, $m = 2$, since $3 \uparrow\uparrow 2 = 27 > 16 = 2 \uparrow\uparrow 3$. The arXiv version of the same paper carries neither the restriction nor the remark about the reader, but its proof of Lemma 4.8 rests on the same Lemma 4.6 and takes the same step, so neither printed proof establishes the base 2. We therefore give a complete proof, in the strict form, for every base $x \geq 2$. It uses only Lemma 2.8 and the facts of Appendix B, and it has been formalised (Section 16).

**Lemma C.1** (Nested upper bound)**.** *Let $r \geq 4$, $x \geq 2$ and $y, z \geq 1$. Then*

$$\mathrm{H}_r\big(\mathrm{H}_r(x, y), z\big) \;<\; \mathrm{H}_r(x, y + z).$$

*In particular* (2) *holds.*

*Proof.* From Appendix B we use that $\mathrm{H}_s(x, n) \geq n+1$ and $\mathrm{H}_s(x, n) \geq xn$ for $s \geq 2$ and $n \geq 1$, and that $\mathrm{H}_s(x, \cdot)$ is increasing. At every rank $s \geq 3$ put $W = \mathrm{H}_s(x, y-1)$ and $T = \mathrm{H}_s(x, y)$, so that $T = \mathrm{H}_{s-1}(x, W)$ and $W \geq 1$ (it is 1 when $y = 1$ and at least $x$ otherwise). The induction is on the rank.

*Rank* 4. Here $T = x^W$, and we show that for every $z \geq 1$

$$W\big(\mathrm{H}_4(T, z) + 1\big) \;\leq\; \mathrm{H}_4(x, y + z); \tag{4}$$

since $W \geq 1$ this gives $\mathrm{H}_4(T, z) < \mathrm{H}_4(x, y + z)$, which is the assertion. For $z = 1$ the two sides are $W(T+1)$ and $x^T$. As $T = x^W \geq W + 1$ we have $W(T+1) \leq (T-1)(T+1) <$

$T^2 = x^{2W}$, and $2W \le 2^W \le x^W = T$ by Lemma 2.8, so $x^{2W} \le x^T$. Suppose (4) holds at $z$, and put $h = \mathrm{H}_4(T, z)$ and $M = \mathrm{H}_4(x, y+z)$, so that $h \ge 1$ and $M \ge Wh + W$. Then $\mathrm{H}_4(T, z+1) = T^h = x^{Wh}$ and $\mathrm{H}_4(x, y+z+1) = x^M$, and since $x^{Wh} \ge x^W \ge W+1$,

$$x^M \;\ge\; x^{Wh}\,x^W \;\ge\; x^{Wh}(W+1) \;\ge\; Wx^{Wh} + W \;=\; W\big(\mathrm{H}_4(T, z+1) + 1\big),$$

which is (4) at $z+1$.

*From rank $r \ge 4$ to rank $r+1$.* Assume the lemma at rank $r$; only its non-strict form will be used. At rank $r+1$, where $T = \mathrm{H}_r(x, W)$, we show that for every $z \ge 1$

$$W + \mathrm{H}_{r+1}(T, z) \;<\; \mathrm{H}_{r+1}(x, y+z), \tag{5}$$

which again gives the assertion because $W \ge 1$. For $z = 1$ the two sides are $W + T$ and $\mathrm{H}_r(x, T)$, and $W + T < 2T \le xT \le \mathrm{H}_r(x, T)$ because $T = \mathrm{H}_r(x, W) \ge W + 1$. Suppose (5) holds at $z$, and put $h = \mathrm{H}_{r+1}(T, z)$ and $M = \mathrm{H}_{r+1}(x, y+z)$, so that $h \ge 2$ and $W + h \le M - 1$. By the lemma at rank $r$ and monotonicity,

$$\mathrm{H}_{r+1}(T, z+1) = \mathrm{H}_r\big(\mathrm{H}_r(x, W), h\big) \;\le\; \mathrm{H}_r(x, W+h) \;\le\; \mathrm{H}_r(x, M-1) =: V,$$

and $V \ge M - 1 \ge W + h > W$. Since $r - 1 \ge 2$ and $V \ge 1$,

$$W + \mathrm{H}_{r+1}(T, z+1) \;\le\; W + V \;<\; 2V \;\le\; xV \;\le\; \mathrm{H}_{r-1}(x, V),$$

and $\mathrm{H}_{r-1}(x, V) = \mathrm{H}_r(x, M) = \mathrm{H}_{r+1}(x, y+z+1)$; this is (5) at $z+1$. □

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## Competing interests

The author has no competing interests to declare that are relevant to the content of this article.

## Author contributions

The paper has a single author, who chose the problem, directed the work, and is responsible for the whole of the manuscript.

## Data and code availability

There is no dataset. The two Lean 4 developments that accompany this article, together with their audit scripts and the raw audit logs, are archived and citable at `https://doi.org/10.5281/zenodo.22789946`; the concept DOI `https://doi.org/10.5281/zenodo.22779441` resolves to the latest version. They are also at `https://github.com/turahigashi/hyper-tamari` under the tag `v1.15.0-pre`, which is the version the archive was cut from and the version Section 16 describes. Reproducing the build requires Lean 4.30.0 and Mathlib 4.30.0 at the revision named in Section 16; the remaining checks need only Python 3.

## Ethics approval

Not applicable.

### Use of AI-assisted tools

Generative AI tools (large language models) were used to find the proofs, and also for the computations, the formalisation and the writing. The author directed the work, reviewed and verified the resulting mathematical arguments, and takes full responsibility for the manuscript.

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